

The 40-year journey for fairness in credit: A systematic review and future research directions

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In this study, we examine the complex, deep-rooted issue of fairness using a technical, historical, and societal lens. We direct our attention to credit scoring as an ideal case study, as it is the subject of a unique standard-setting process with a remarkably long history of dealing with issues of bias and discrimination. While previous works have assessed fair practices via generalizable statistical means, few have considered domain-specific, systemic factors specific to the target application. This is crucial for solving the theory-to-practice gap which remains largely unaddressed. We conduct a systematic literature review including descriptive and thematic analyses of 185 articles from 1978 to 2022, highlighting the scale and evolution of the extant body of knowledge. The study proposes an integrated systemization of the knowledge structure across six themes: 1) identification, 2) analysis, 3) transition, 4) optimization, 5) performance and 6) application. We find that scholarly attention is directed at multiple lines of study, including identifying disparity and evaluating its causal factors, assessing technology-driven shifts and novel risks associated with complex machine learning algorithms, investigating sources of bias across the lending process and its relationship to environmental factors (i.e., policy, social context, business market), and ultimately determining how these amount to forming the causes, consequences, and solutions for achieving fair outcomes. We highlight trends over time, providing insight into problematic or neglected areas. We conclude with a discussion on remaining gaps and suggestions for a more productive route towards empirical and practical development of fair credit scoring approaches.

KEYWORDS

Artificial intelligence, Fairness, Discrimination, Machine learning, Credit scoring, Financial Technology (FinTech)

1 | INTRODUCTION

Credit is an essential component to financial wellbeing; therefore, unequal access can further strain the economic disparities seen across demographic groups today. However, with increasingly complex machine learning (ML) algorithms being used as part of automated decision-making (ADM) processes, there is concern that algorithmic bias and unfairness may be encoded and perpetuated via these systems. This poses a challenge to lenders as well as regulators, who are faced with the crucial task of updating their enforcement practices and issuing new guidance to avert AI-induced harms. Most recent conversations around fair ML are dominated by computer scientists; however, findings are often presented in abstract, application-agnostic terms. Meanwhile, without expert guidance and insider knowledge from lenders themselves, a clear picture is lacking as to which tools or metrics are useful, legal, and most importantly, practical for resolving the fairness issue. In other words, there is a major gap between the existing knowledge and efforts which are application (credit)-specific and technology (sophisticated ML)-specific.

From the literature, it is evident that credit scoring serves as a common example in algorithmic bias work, from which parallels to other domains using AI can be drawn. Automated credit scoring is a well-studied field with origins dating back to post-World War II when consumer credit ballooned [1]. Consequently, lending has a long and rich history looking at issues of bias and discrimination and has been the subject of a unique standard-setting process amidst a myriad of changing environmental factors. Its popularity can also be attributed to the availability of public benchmarking datasets sourced from this field; however, these datasets are discovered to be a contributing factor to the outdated, impractical nature of recently proposed fairness techniques.

The main reason for undertaking this systematic literature review (SLR) is to consolidate the diverse literature in this context across multiple disciplines to provide a comprehensive review of fair credit research. We map the field's evolution, identify distinct research streams, and uncover shifts in methods used, systematizing them into a framework for applicability to other areas in society where algorithmic bias is a concern. Notably, despite the burgeoning work on fairness and bias in the last decade [2], attempts to reconcile these works with previous empirical studies on lending discrimination remain sparse.

The contributions of this paper can be summarized on two levels. First on a more general level, we contribute to the body of research on responsible and ethical AI. By reviewing a specific application of algorithmic decision-making systems and historical data examples, we add to the wider knowledge of understanding what fairness values are, the needs of its relevant stakeholders, how they can be achieved, and what gaps remain. In addition to novel content, we also provide an element of methodological innovation. By formulating different domains of knowledge, we synthesize a coherent, systematic framework to enhance the existing methodology which can be used in future reviews on credit scoring as well as a broader investigation of fairness.

On a context-specific level, we make several important contributions to the credit scoring literature. Firstly, this paper builds on and extends previous reviews which focus on only one or two of the themes delineated in this study, credit, fairness and machine learning [3, 4, 5]. One theme often falls out of scope even though they are related, thus there is limited interaction between these research streams resulting in a fragmented body of knowledge. In doing so, we also provide original insights into the transformative nature of the research profile, plotting an evolution from identifying cases of discrimination via data-centric analyses to emerging trends around fairness optimization led by computer scientists via algorithm-centric analyses. To better understand how to secure fair outcomes, we then trace sources of discrimination along the lending process and investigate the 'why' behind observed, real-world disparities and whether they are justifiable. Notably, we highlight factors involved pre-application (e.g., financial inclusion, applicant outreach), during (e.g., lender-lendee homophily, institutional characteristics) and post-application (e.g., default, delinquency, and repayment behavior). Furthermore, scholars agree that state-of-the-art ML models harbor novel

risks, therefore we also link algorithmic drivers of discrimination (e.g., data bias, triangulation, missingness), their consequences and how they eventually fare from the perspective of those impacted. Finally, we identify remaining gaps, categorize fertile avenues for future research and discuss the implications, both theoretical and practical.

2 | METHODOLOGY

Inspired by previous work in the field of SLRs ([6, 7, 8]), we amend and localize stages according to our conditions and define a three-phase process.

2.1 | Stage one: Question formulation

A careful formulation and clear design of research questions (RQ) is a fundamental step for ensuring accuracy and validity of later stages. We define three RQs to guide us:

(RQ1) How have key studies in fair credit decision-making evolved over time, and what are the underlying research streams?

(RQ2) What identification, measurement and repair mechanisms have been used in the literature to combat discrimination and unfairness in credit decision-making systems?

(RQ3) Can fair credit practices and modeling measures be organized into an integrated systemization, from which we can derive the lessons learned and problems that remain unresolved?

2.2 | Stage two: Search and filtration

Next, we establish search boundaries, identify search strings, and specify time frames for selecting studies. The initial search was conducted in November 2022 with no time limitations to capture the full historical development of the scholarly body. Documents are gathered from two widely employed databases, Scopus and Web of Science (WOS) (see full flowchart in Figure 1).

To begin with, we search both databases for titles, abstracts, and keywords containing terms for two major themes – fairness and credit (Table 1). We use truncation to find different variants of relevant terms, the Boolean operator OR for multiple terms within a theme, and operator AND for searching across themes. The initial search yields 2,725 and 2,835 articles from Scopus and WOS, respectively. Given the large number of preliminary results, we limit our search to articles written in English and located within disciplines of business, management and accounting, computer science and engineering, decision sciences, economics, and social sciences, which make up the majority. Removing duplicates results in a merged sample of 2,926 potentially relevant documents. In the second stage, articles are screened to meet the inclusion/exclusion criteria defined in Table 2 to ensure relevance. Next, accepted papers are scanned based on full text review and 66 are eliminated.

2.3 | Stage three: Extraction, analysis, and synthesis

We conduct two kinds of analysis on our final sample – descriptive and thematic. The descriptive analysis is used to identify the distribution of articles based on publication year, outlet, geography, and methodology. We also use a co-occurrence network to bring a sense of order to the various research streams. However, because common variables can indicate agreement or disagreement, understanding article content is required. A thematic analysis is undertaken,

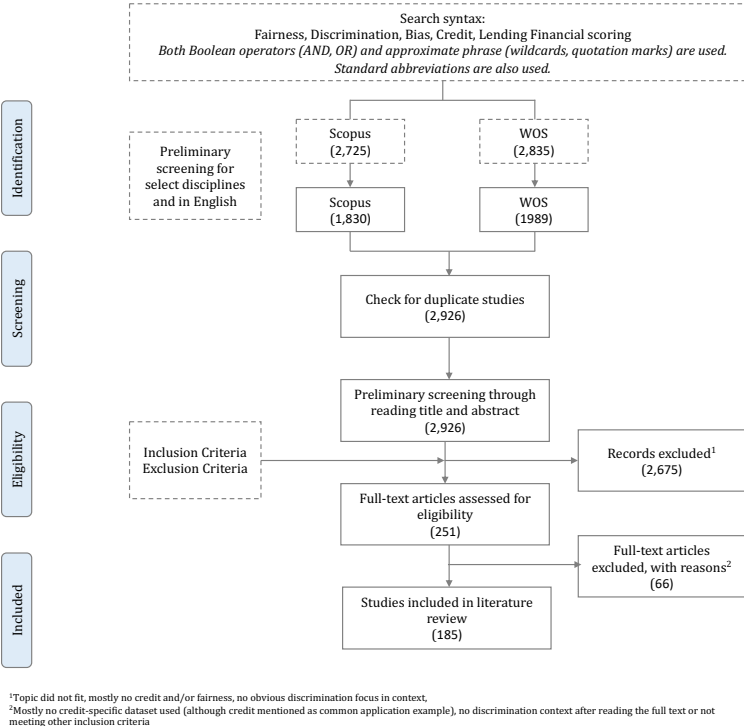


FIGURE 1 PRISMA flow diagram illustrating the search protocol.

involving reading articles in full and using a data extraction base to encode relevant items. This enables us to map methodological patterns and develop an evolutionary record. In doing so, we identify the pertinent literature around fairness in credit and the boundaries of sub-themes, e.g., how disruptive technologies can affect traditional credit scoring methods.

3 | DESCRIPTIVE ANALYSIS

In this section, we address our first research objective, i.e., to identify the level of extant research by describing key bibliographic dimensions. The descriptive analysis provides an important backdrop for informing the thematic categorizations in the subsequent content analysis.

3.1 | Temporal analysis

The final sample consists of 185 articles drawn from 129 journals and conference proceedings between 1978 and 2022. Although the first relevant publication appeared in 1978, the field did not attract much attention until the early 21st century. Figure 2 displays publications per year, with an annual growth rate of 7.8%. Our study has no time delimiters, however, more than half (54%) of the articles were published in the last five years, demonstrating a relatively recent boost of interest from the scientific community. This coincides with Larsson's [9] claim that AI

Component 1		Component 2
('fair*' OR 'unfair*' OR 'bias measure' OR 'bias mitigation' OR 'bias reduction' OR 'bias identification' OR 'discriminatory' OR 'discrimination' OR 'sensitive feature' OR 'sensitive attribute' OR 'protected characteristic')	AND	('credit risk' OR 'financial risk' OR 'credit scor*' OR 'financial scor*' OR 'consumer scor*' OR 'digital scor*' OR 'loan*' OR 'lending' OR 'credit card' OR 'mortgage*' OR 'non-payment').

TABLE 1 Search string for SLR.

Inclusion criteria	i.	offers empirical research relating to discrimination and/or fairness in credit, including quantitative, experimental, analytical or case studies,
	ii.	where the identification, analysis, or mitigation of fairness and/or discrimination in the credit context is the main focal point of the paper and not a supplementary aim; for example, a paper investigating the optimization of credit scoring models for improving default prediction where fairness is a supplementary check is excluded,
	iii.	conducted within the context of a credit scoring model or decision-making system relating to a lending process,
	iv.	utilizes a credit-related dataset in the analysis, including work with personnel such as loan officers or applicants.
Exclusion criteria	i.	non-empirical papers,
	ii.	papers that assess non-individual credit decisions (e.g., small businesses and SMEs) given the ambiguity between discrimination against the individual versus the financial capability of the business,
	iii.	non-academic articles (e.g., editorials, extended abstracts, book chapters)

TABLE 2 Inclusion and exclusion criteria for SLR.

publications combined with ethics and governance have grown remarkably over the last two to five years. We also feel the trend can be attributed to consumer pressure and recent regulatory events, which have encouraged greater public discussion on the topic.

For example, the White House released a report on algorithmic systems calling for “equal opportunity by design” as a guiding principle in domains like credit scoring [10]. A UK House of Lords report by the Select Committee on AI [11] summarized difficulties with developing un-biased AI quoting, “More needs to be done to ensure that data is truly representative of diverse populations and does not further perpetuate social inequalities.” In 2019, the EU Commission’s High-Level Expert Group on AI [12] published ethical guidelines stating seven key requirements for realizing ‘trustworthy AI’ with ‘diversity, non-discrimination and fairness’ being one. In response, the first steps toward an international framework for AI governance were made (‘OECD/LEGAL/0449’, 2019). In 2021, another significant event took place – the EU’s AI Act was proposed, together with the AI Liability Directive, which affords the right to sue companies if a subject is harmed by an automated system [13]. Similarly, the Biden-Harris administration released a “Blueprint for an AI Bill of Rights” looking to guide the design, development, and deployment of automated systems [14] and developing legislation would require companies to conduct impact assessments for bias and create, for the first time, a public repository of such systems [15].

3.2 | Distribution of articles by journal and discipline

We notice that business, finance and economics scholars, so far, conducted the highest number of studies in this space (Tables A.1, A.2). However, as ML technologies began monopolizing classification tasks, ‘algorithmic bias’ has recently been absorbed by computing disciplines. Even AI ethics studies represent a small but growing part of the base. The

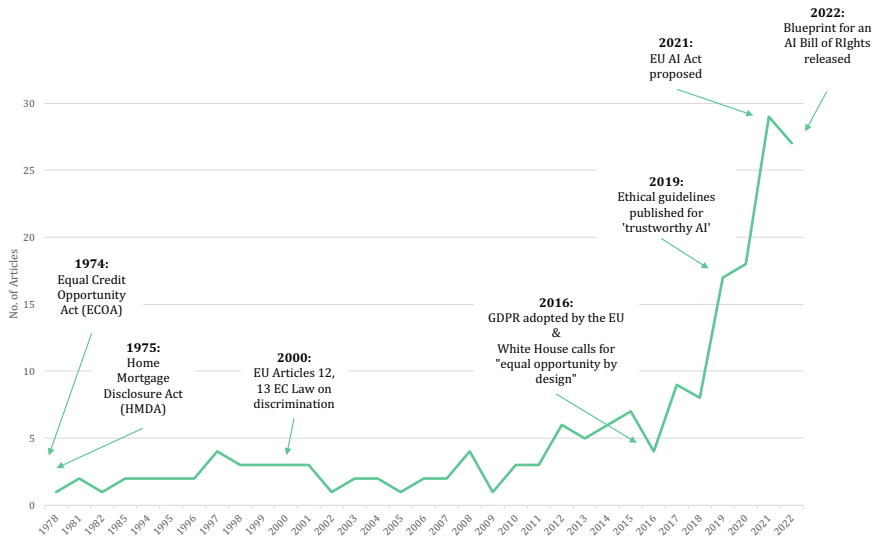


FIGURE 2 Publications over time. Major regulatory and discrimination policy-related events are highlighted.

disparity of outlets is indicative of the diversity of works in the 'fairness in credit' literature. While this promises novel insights via a cross-disciplinary analysis, it also hints at the fragmented nature of the research community.

3.3 | Citation Impact

Table 3 presents the top 15 most cited studies. Although they represent less than a fifth of the sample size, they attract 55% of all citations. However, because citations grow over time, it is important to note that five of these studies were published before 2000. The article by Munnell et al. [16] ($n=488$) can be described as seminal to the growth of the field. The next most cited are Pope and Sydnor [17] and Hajian and Domingo-Ferrer [18]. We also observe a focal shift towards discrimination in the context of ML and financial technologies (FinTech), as opposed to more traditional methodologies (i.e., statistical scoring) and applications (i.e., mortgages).

3.4 | Geographic analysis of authorship origin

Overall, US-based scholars have dominated this topic ($n=115$), followed by contributions from Europe ($n=43$) (Figure A.1). This may mirror countries' public and private investments into digital technologies [4] or socially liberal outlooks with regards to egalitarian views (e.g., gender roles) also reflected in their legal constructs.¹ With regulations in place, policy makers and publicly funded bodies may drive more investment in this direction relative to countries where these societal issues are of less concern [3]. Notably, areas with more conservative outlooks (e.g., skewed gender roles) or social hierarchies (e.g., caste system) could provide interesting insight in contrast to US-oriented perspectives

¹The US and EU countries enforce strict anti-discrimination laws; for example, the Fair Credit Reporting Act, Equal Credit Opportunity Act and the Fair Housing Act in the US and consumer protection directives like Articles 12 and 13 EC and the General Data Protection Regulation in the EU. These foundational pieces of legislation prohibit the use of protected characteristics such as national origin, race, religion, sex, age, religion or sexual orientation as basis for decision-making.

where strength of racial tensions, for instance, dominate.

Author (Year)	Title	Journal	Citations (average citations / year)	Country
Munnell A.H. (1996)	Mortgage Lending in Boston: Interpreting HMDA Data	American Economic Review	488 (18.1)	USA
Pope D.G. (2011)	What's in a picture?: Evidence of discrimination from Prosper.com	Journal of Human Resources	346 (28.8)	USA
Hajian S. (2013)	A methodology for direct and indirect discrimination prevention in data mining	IEEE Transactions on Knowledge and Data Engineering	145 (14.5)	Spain
Agier I. (2013)	Microfinance and Gender: Is There a Glass Ceiling on Loan Size?	World Development	118 (11.8)	Belgium
Bellamy R.K.E. (2019)	AI Fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias	IBM Journal of Research and Development	117 (29.3)	USA
Barasinska N. (2014)	Is Crowdfunding Different? Evidence on the Relation between Gender and Funding Success from a German Peer-to-Peer Lending Platform	German Economic Review	108 (12.0)	Germany
Ruggieri S. (2010)	Data mining for discrimination discovery	ACM Transactions on Knowledge Discovery from Data	103 (7.9)	Italy
Ghosh S. (2017)	What Constrains Financial Inclusion for Women? Evidence from Indian Micro data	World Development	91 (15.2)	India
Berkovec J.A. (1994)	Discrimination, competition, and loan performance in FHA mortgage lending	Review of Economics and Statistics	87 (3.0)	USA
Bocian D.G. (2008)	Race, ethnicity and subprime home loan pricing	Journal of Economics and Business	82 (5.5)	USA
Holloway S.R. (1998)	Exploring the neighborhood contingency of race discrimination in mortgage lending in Columbus, Ohio	Annals of the Association of American Geographers	81 (3.2)	USA
Aggarwal R. (2015)	Lending to women in microfinance: Role of social trust	International Business Review	71 (8.9)	USA
Chen J. (2019)	Fairness under unawareness: Assessing disparity when protected class is unobserved	Proceedings of the 2019 Conference on Fairness, Accountability, and Transparency	63 (15.8)	USA
Berkovec J.A. (1998)	Discrimination, competition, and loan performance in FHA mortgage lending	Review of Economics and Statistics	56 (2.2)	USA
Black B.H. (1978)	Discrimination in mortgage lending	American Economic Review	54 (1.2)	USA

TABLE 3 Articles with the greatest impact based on number of citations.

3.5 | Field work characteristics

3.5.1 | Methodology and dataset coverage

We categorize works in terms of (1) self-reported data using surveys, interviews, workshops, and human-subject experiments, (2) observational data using tabular datasets, and (3) mixed methods which employ both. Observational data methods dominate (n=162), followed by self-reported data (n=23). Our review highlights the overwhelming usage of two secondary datasets (Figure 3), HMDA (n=48) and German credit (n=24), in particular. HMDA is the most comprehensive dataset on US mortgage market activity [19] given its large representation, annual updates and inclusion of race/ethnicity, which is seldom collected due to privacy violations. We find that German credit is often used alongside other datasets unrelated to credit; 21 out of 24 studies employed German credit alongside datasets like Adult census and COMPAS recidivism [20, 21], which are common benchmarks in fair ML work. Here, the contribution is an algorithmic framework (e.g., regularization approach) as opposed to a credit-specific solution.

3.5.2 | Geographic and protected characteristic coverage

To identify the economies which have received greater focus, the country from which the dataset was sourced is considered. Not surprisingly, the most studied countries are the US (n=104), Germany (n=29), China (n=8), India (n=7), and UK (n=6). This can be attributed to the geographic distribution of authors (Figure A.1) and the fact that HMDA is collected by the US government. Countries like China and India are also prominent, reflecting their economic growth

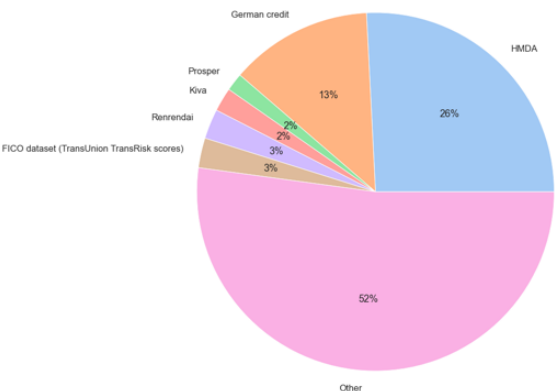


FIGURE 3 Distribution of popular datasets used for analysis.

and recent interest in the topic, particularly with regards to peer-to-leer (P2P) lending and the salience of gender (Chen, Li and Lai, 2017) as well as caste-based discrimination [22, 23]. While most computer science publications are generic in terms of target protected characteristic (PC), some focus on a specific vulnerable group. Depending on the credit application, certain PCs are more frequently considered (Table 4): race/ethnicity in mortgage lending, gender in consumer credit and age and education level in P2P lending.

3.5.3 | Credit application coverage

‘Credit’ is a broad term applied across different financial institutions and products (Figure A.2). Nearly half (n=80) investigate mortgages, followed by general bank loans (n=25) with those ‘not applicable’ using credit as one of several domain examples.

3.6 | Keywords co-occurrence

A science mapping is applied to help identify the relationship (interactions or connections) between research constituents [24]. The co-occurrence of keywords analysis assumes that keywords that appear simultaneously and frequently are connected via a thematic relationship [25]. We use the software tool VOSviewer [26] to visualize prominent themes (Figure 4, see Appendix for details). These occurrences signify the influential role of technology, credit application and PC when understanding a multi-faceted concept like fairness. Therefore, when any of these factors differ – *who* is being discriminated against, via what *application* harms are occurring, or what *model* or *human* is deployed – the impact on fairness and necessary considerations differ. Moreover, a dynamic version (Figure A.3) illustrates the recent technology-oriented shift in academic interest.

Sensitive Attribute									
	Ethnicity / Race	Gender	Nationality / Immigra- tion status	Caste	Sexual orientation / Marriage status	Age	Occupation	Education	Children
Authorship Origin									
Australia				1					
Belgium	1	4			1	1			
Canada	1	1							
China	2	3	1			1		3	
Denmark			1						
France	2	2					1		
Germany		1	1						
India		1		1					
Italy		3	2						
Japan	2	2				2			
Mexico	1								
Nigeria		1							
Norway	1	1							
Poland									
Qatar	1	1							
South Africa									
South Korea	1	1				1			
Spain			1						
Switzerland	1								
Taiwan	1	2							
UK	4	5		2					
USA	80	19			5	8		1	1
No. of Studies	98	47	6	4	6	13	1	4	1

TABLE 4 Distribution of PCs studied by authorship origin.

4 | THEMATIC ANALYSIS

We categorize extant research into six themes: 1) identification, 2) analysis, 3) transition, 4) optimization, 5) performance and 6) application. This helps clarify boundaries of sub-areas and their differentiating focal points. We synthesize a framework (Figure 5) which outlines the overall knowledge structure (see Discussion for more details) and dive deeper into each theme.

4.1 | Identification

4.1.1 | Defining discrimination

Discrimination-aware studies typically utilize conventional statistical approaches to measure bias, selecting a dependent variable such as rejection, pricing or default to independent variables such as applicant characteristics, product (e.g., property, loan terms) and economic environment. Researchers have noted significant disparities in lending rates among PC-defined groups – two competing explanations arise.

One argument is that outcome disparities result from underlying disparities in creditworthiness. Lenders argue that factors like income rather than race are the cause; consequently, if minorities as a group are poorer than Whites, this would explain their higher rejection rates [27]. Using Freddie Mac's 1999 Consumer Credit Survey, researchers also found that much of the racial gap in denials can be explained by racial disparity in credit scores [28]; however, researchers have highlighted the disparate impact of credit scoring [29, 30, 31]). One can argue that racial differences in credit risk are seldom independent of historically discriminatory policies and practices. Rather, their legacy may continue to manifest itself in today's observations; therefore, while creditworthiness may "explain" lending gaps, there is still the task of understanding why and how disparities manifest in the first place.

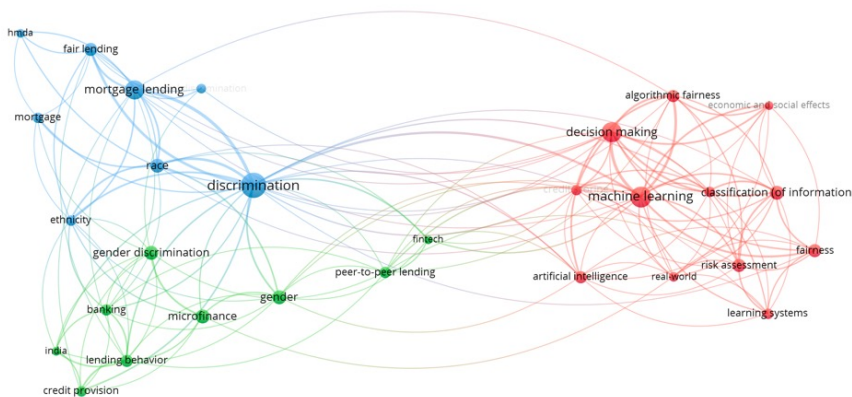


FIGURE 4 Keyword co-occurrence analysis using VOSviewer.

A competing view is that the disparities arise from lenders themselves. A key point of contention in this debate is the distinction between disparate treatment (direct discrimination) and disparate impact (indirect discrimination). Disparate treatment occurs when lenders apply different or inconsistent treatment based on prohibited criteria which are not fully explained by justified and non-discriminatory factors. Disparate impact occurs when applicants receive equivalent, neutral treatment, however, the policy has a burdening effect regardless of the absence of discriminatory intent. This tends to be unplanned as they are a consequence of consistent policies and their application, however, in many instances is difficult, if not impossible, to disentangle the two.

4.1.2 | Testing for discrimination

The first example of discrimination testing is found in Black et al. [32] who use a large-scale survey of banks and borrowers on mortgage applicants, including outcome, terms, financials, and demographics. Using a single equation model, they were one of the first to present evidence that race and age have a significant positive correlation with rejection. This served as basis for testing discrimination using a conventional approach of rejection equations. Shortly after, the Boston Federal Reserve published a pioneering study which confirmed the presence of racial discrimination in mortgage lending [16]. This seminal study used 1991 HMDA data accounting for variables linked to default risk, cost of default, loan characteristics, and personal and neighborhood characteristics of over 3000 applicants and 100 financial institutions. It concluded that Black and Hispanic applications were 82% more likely to be rejected than similar White applications. It stood out as one of the most prominent studies on credit discrimination, evidenced by its high citations (Table 3), supporting the view that rejection rate disparity was attributable to unequal treatment of those equally qualified.

Unsurprisingly, these findings caused significant consternation and outcry by national regulators, lenders, and public; they stood in stark contrast to what the industry had been preaching. Researchers began scrutinizing the study to discredit its findings; critics argued the study was flawed based on data quality [33], model specification problems [34, 35], omitted variable bias [36] and endogenous explanatory variables, such as that Blacks are more likely to default. In other words, if discrimination against Blacks existed, only the very “best risks” would access loans, therefore, only the “best risks” among Blacks would be observed in default [28]. This would require their default rates to be lower than average [37]; however, because this is not the case, Blacks are not being discriminated against

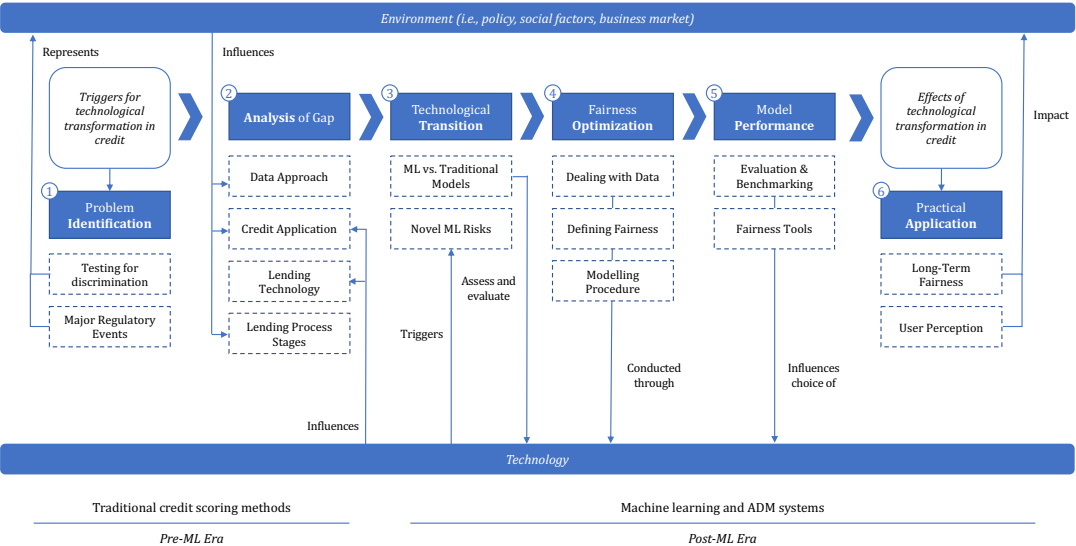


FIGURE 5 An integrative process framework for fair credit research.

[38, 39].

In the face of skepticism, many also defended the notion that discrimination plays a major role in outcomes [40, 41, 42, 43, 27, 44, 45]. Strong counter-findings supported the original Boston Fed results, where one detailed replication found “an African American is still more than twice as likely as a White to have his home mortgage loan application rejected [46].” Consequently, similar analyses were applied to areas outside of Boston to determine if trends were ubiquitous. One analysis in Milwaukee, Wisconsin [47] found that racial effects against Black and Latino customers persisted after controlling for several financial, housing, and neighborhood characteristics. Similar studies were conducted in New Jersey [42], Ohio [48], Indiana [49], Kansas [27], and Los Angeles [50, 51], which included spatial patterns to identify lender bias against poorer/minority-concentrated neighborhoods, and in Washington D.C. [52], which included interviews with bank officers. Table A.3 displays how these studies are distributed by dataset, time period, and outlet. The application of mortgage loans dominated the literature in its early, formative years, with race/ethnicity being often the target characteristic of the analysis [53, 54, 55, 56, 57, 58].

4.2 | Analysis

We have seen evidence of the inescapable conclusion: “Race was indeed a fairly powerful influence in lending decisions [59][p.15].” The challenge faced by researchers now is uncovering more covert forms of lending behavior that can systematically burden marginalized groups. This can manifest in several ways depending on the context (Figure 6). We begin by surveying the methods adopted for data collection and analysis (Table 5).

4.2.1 | Data Approach

According to statistical principles of design, we first classify methodologies based on how data is collected. Sources are categorized based on level of control – having no control over independent variables (i.e., observational), control

Level	Phase in credit application	Impact considered	Authors (Year)	Credit Application	Protected characteristic(s) of interest	Data Used	Country
Consumer	Pre-application & Access	Borrower discouragement	Caglayan M. (2022)	Refinancing	Gender	Renrendai	China
	Pre-application & Access	Borrower outreach	Nyarko S.A. (2022)	Microfinance	Gender	Rating reports of five leading microfinance rating agencies	Multiple
	Pre-application & Access	Financial inclusion	Rao S. (2015)	Financial inclusion	Gender; Child	US Survey of Consumer Finance data	US
	Pre-application & Access	Financial inclusion	Ghosh S. (2017)	Financial inclusion	Gender	All India Debt and Investment Survey	India
	Pre-application & Access	Financial inclusion	Bullock H.E. (2020)	Financial inclusion	Race / ethnicity; Gender	108 low-income Latinx mothers	US
	Pre-application & Access	Financial inclusion	Afolabi A.O. (2021)	Financial inclusion	Gender	Selected loan officers	Nigeria
	Credit outcome	Education level	Li J.W. (2019)	P2P lending	Education	Renrendai	China
	Credit outcome	Education level	Huang W.L. (2020)	General loan	Education	Xinsindai	China
	Credit outcome	Female partner in labor market	Robinson J.K. (2002)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Lender search effort	Cheng P. (2011)	Mortgage loan	Gender	Survey of Consumer Finances	US
	Credit outcome	Length of credit history	Han S. (2011)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Self-perception of creditworthiness	Ards S.D. (2015)	General loan	Race / ethnicity	Freddie Mac's Consumer Credit Survey	US
Lender	Pre-application & Access	Lender subjectivity	Hanson A. (2016)	Mortgage loan	Race / ethnicity	Matched pair correspondence experiment using e-mail inquiry	US
	Pre-application & Access	Lender subjectivity	Hernández-Trillo F. (2022)	General loan	Race / ethnicity	Three professional actors	Mexico
	Credit outcome	Lender-Lendee Homophily	Squires G.D. (1995)	Mortgage loan	Race / ethnicity	DA data, EEO-1 data, and 1990 U.S. Census data	US
	Credit outcome	Lender-Lendee Homophily	Black (1997)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Lender-Lendee Homophily	Black H.A. (2001)	Mortgage loan	Race / ethnicity	HMDA, Federal Reserve Board's National Information Center, Call	US
	Credit outcome	Lender-Lendee Homophily	Mengoli S. (2017)	Microfinance	Gender	205 MFIs in 54 countries	Multiple
	Credit outcome	Lender-Lendee Homophily	Beck (2018)	Microfinance	Gender	Albanian lender	Albania
	Credit outcome	Lender-Lendee Homophily	Chen J.I. (2019)	Microfinance	Race / ethnicity; Gender	Undergraduate students at Brown University	US
	Credit outcome	Lender-Lendee Homophily	Goenner (2022)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Lender subjectivity	Pope D.G. (2011)	P2P lending	Race / ethnicity	Prosper marketplace	US
	Credit outcome	Lender subjectivity	Agier (2013)	Microfinance	Gender	VivaCred	Brazil
	Credit outcome	Lender subjectivity	Hossein C.S. (2014)	Microfinance	Race / ethnicity	93 interviews . Case study	Guyana
	Credit outcome	Lender subjectivity	Harkness (2016)	P2P lending	Race / ethnicity; Gender	Amazon's Mechanical Turk participants	Multiple
	Credit outcome	Level of information sharing	Alharato G. (2022)	General loan	Nationality / immigration status	Credit Register by Banca d'Italia	Italy
	Credit outcome	Social trust	Aggarwal R. (2015)	Microfinance	Gender	Microfinance Information Exchange	Multiple
	Credit outcome	Social impact	Dorflteiner G. (2021)	P2P lending	Nationality / immigration status	Kiva.org	US
	Post-application perception	Lender-Lendee Homophily	Kulik C.T. (2000)	General loan	Race / ethnicity; Gender	Survey of loan applicants	US
Institutional	Credit outcome	Model specification	Horne (1997)	Mortgage loan	Race / ethnicity	Boston Fed data	US
	Credit outcome	Bank practices and policies (institution type, local/nonlocal ownership, bank size, locations of bank branches)	Williams R.A. (1997)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Lender characteristics (risk tolerance/preference, bank size, institution type, profitability)	Harrison D.M. (2001)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Bank-level vs. market-level model	Blackburn M. (2006)	Mortgage loan	Race / ethnicity	OCC examinations	US
	Credit outcome	Lender's ex post foreclosure risk	Bayer (2018)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Incentivization (short-term outcome vs. long-term profits)	Dobbie W. (2021)	General loan	Race / ethnicity; Age	High-cost UK lender	UK
	Credit outcome	Prosocial service orientation framework	Meshram K. (2022)	Financial inclusion	Caste	Loan officers and lower caste loan applicants across 43 MFIs	India
	Credit outcome	Lender's loan guarantee policy	Stroube B.K. (2022)	P2P lending	Gender	Unnamed P2P platform in China	China
	Credit pricing	Rate declines	Crawford G.W. (1999)	Mortgage loan	Race / ethnicity	Conventional, FHA and VA loans from City Federal Savings Bank of New Jersey	US
	Credit pricing	Overages	Black H.A. (2003)	Mortgage loan	Race / ethnicity	Undisclosed major mortgage lending institution	US
In-application	Credit pricing	Credit scoring	Avery R.B. (2012)	Credit scoring	Race / ethnicity; Gender; Age	Nationally representative sample of credit bureau records	US
	Credit pricing	Credit scoring	Bernerth J.B. (2012)	Credit scoring	Race / ethnicity; Age; Education	University students	US
	Credit pricing	Ceiling	Cozarenco A. (2018)	Microfinance	Gender	French MFI	France
	Credit pricing	Discount points	Bhutta N. (2021)	Mortgage loan	Race / ethnicity	FHA-HMDA datas et merged with Optimal Blue	US
	Credit pricing	Yield spread premiums	Collins M.C. (2021)	Mortgage loan	Race / ethnicity	New York Attorney General's office	US
	Credit repayment	Default rates	Berkovec J.A. (1994)	Mortgage loan	Race / ethnicity	FHA-insured mortgages	US
	Credit repayment	Delinquency	Lee J. (2012)	Mortgage loan	Race / ethnicity	Survey of Consumer Finances	US
	Credit repayment	Delinquency	Anyamele O.D. (2015)	General loan	Race / ethnicity	Survey of Consumer Finances	US
	Credit repayment	Default rates	Li J. (2022)	P2P lending	Nationality / immigration status	Renrendai	China
Environmental Impact	Credit outcome	Competition of lending environment	Berkovec J.A. (1998)	Mortgage loan	Race / ethnicity	FHA-insured mortgages	US
	Credit outcome	City size	Coffey B. (1999)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Market forces	Harvey K.D. (2001)	Mortgage loan	Race / ethnicity	HMDA, Bank performance data	US
	Credit outcome	Government policy - Affirmative action	Kumar S.M. (2013)	General loan	Caste	All-India Debt and Investment Survey	India
	Credit outcome	House price inflation	Wheeler C.H. (2015)	Mortgage loan	Race / ethnicity	HMDA	US
	Credit outcome	Riot exposure	Fisman R. (2020)	General loan	Race / ethnicity	Large public sector Indian bank, Hindu-Muslim violence data	India
	Credit outcome	Recession	Deku S.Y. (2022)	Mortgage loan	Race / ethnicity	Living Costs and Food Survey	UK
	Credit outcome	Interstate variation	Antoniades (2017)	Mortgage loan	Race / ethnicity; Gender	HMDA	US
	Credit outcome	Sexual orientation discrimination laws	Dillibary J.S. (2019)	Mortgage loan	Marriage status / sexual orientation	HMDA, Loan data from GMAC	US
	Credit outcome	Mortgage financing regulation	Kau J.B. (2019)	Mortgage loan	Race / ethnicity	HMDA, Loan data from GMAC Residential Capital Company	US

TABLE 5 Causes of discrimination in the reviewed sample, including level of impact, approach and geographical distribution.

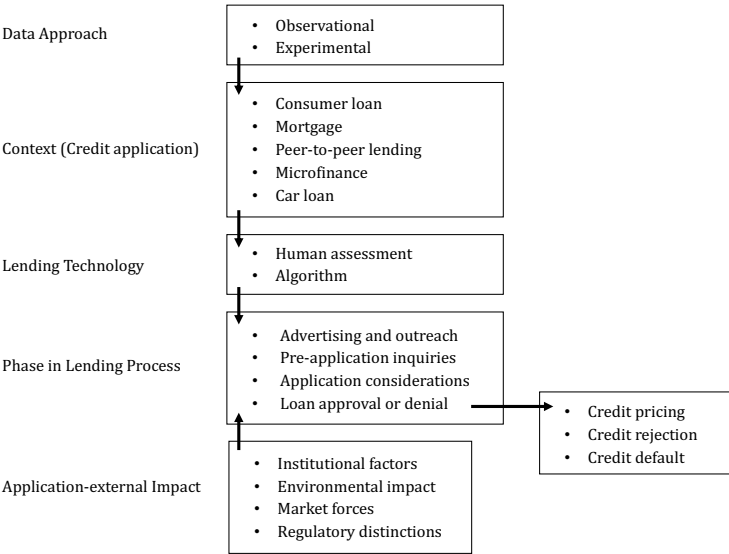


FIGURE 6 Key considerations of an application process that affects lending behavior.

over a few (i.e., quasi-observational) or control over all of them (i.e., experimental). Seeking to bring more realism to their investigations, researchers carefully select, match, and train testers to play the role of a credit seeker.

By presenting equally qualified individuals who differ only by PC, researchers can assess the degree to which PCs affect opportunities. In a laboratory-based experiment by Chen et al. [60] to study willingness to lend, authors found that lenders preferred borrowers with whom they share gender and ethnic similarity even after controlling for perceived riskiness and neediness. An e-mail correspondence experiment by Hanson et al. [61] documented the effect of being Black on mortgage loan originators (MLO) response rate as being equivalent to having a credit score 71 points lower. Most studies are observational where bank records are shared by a data partner; however, this can lead to contention regarding its reliability, completeness of relevant variables, and absence of confounding variables [62].

4.2.2 | Credit application

We further categorize methods based on its context, or credit application. While earlier works focused on mortgage lending – largely due to data availability – at the turn of the century, investigations expanded as new products were offered and digitized data exploded. Financing options include car loans [63], farm loans [64, 65, 66], and P2P lending [67, 68, 69]. P2P lending has particularly grown in the last decade as consumers and investors began to shift away from traditional institutions following the financial crisis and growth of the online space [70]. This resulted in available data from sites like Prosper [17] and Renrendai [71, 72] and is reflected in the recent uptick of studies [73, 74].

Studies also begin to focus on PCs beyond race/ethnicity. For instance, disparities in used car loans are found against minorities and females. These include higher administration costs and default rates [63], lower approval rates and 70-basis-point higher interest in Black and Hispanic applicants [75], as well as taste-based discrimination against female applicants [76]. Researchers find substantial differentials in loan amounts between racial and gender classes for farm loans [65, 66] as well as significant inter-caste differences in application rates and discrimination against

Scheduled Tribe borrowers (i.e., lower caste) in India [23]. In P2P lending, gender is a feature of interest alongside education [68], with heavier investment in Asia, particularly China with Renrendai.² Females are more likely to be funded than males, however, pay the price via higher interest rates in exchange for credit access even with better loan performance [73, 68]. On the contrary, Barasinska and Schäfer [67] found female discrimination eased by the “wisdom of the lending crowd” and, in the Korean P2P market, non-existent [69].

4.2.3 | Lending technology

Different methodologies are used for discrimination identification and analysis depending on the lending technology used. In any decision making process, asymmetric information is a crucial problem. To overcome this obstacle, lenders typically combine two strategies: credit scoring and relationship lending. We characterize this as algorithmic vs. human decision making, respectively. In the former, lenders systematically assess creditworthiness from personal characteristics and financial information [77], whereas the latter is a more time-consuming process by which an officer must learn about their creditworthiness [78] but holds more power over individual cases. To test discriminatory consequences, researchers apply statistical inference to the collected data. Two major theoretical models of economic discrimination are known: taste-based and statistical [79]. The former has no economic basis but originates in subjective prejudice or stereotyping, whereas the latter assumes statistical thinking, however, occurs from a scarcity of information on true risk during application. Due to imperfect information exchange, lenders are incentivized to use more easily identifiable characteristics like race as a proxy for creditworthiness based on prior knowledge of the group's average creditworthiness. Consequently, applicants from disadvantaged groups may experience lending differentials. However, unlike taste-based discrimination, an increase in applicant information often results in lower disparity [80].

In microfinance, due to limited use of technology, relatively cheap human capital [81] and demand far exceeding supply [82], credit officers have greater flexibility to act on their own judgement. Thus, arbitrary choices and freedom to act on historically rooted prejudices can result in discriminatory patterns via selective utilization of race-neutral standards [83]. Agier and Szafarz [81] found gender affected lenders' decisions regarding loan size by altering their status beliefs. These results were obtained using partial-least-square estimation to mimic the loan allocation process using data from VivaCred, a Brazilian microfinance institution (MFI). Disentangling the objective assessment of creditworthiness and subjective judgement of an officer requires dissecting the underlying decision mechanism. However, due to the difficulty and insider knowledge needed, some researchers utilize experimental methods. Using Amazon Mechanical Turk participants in an artificial P2P lending market, Harkness [84] found gender and race significantly affected funding decisions.

Credit scoring, on the other hand, is a method commonly used in mainstream banking that assigns a numerical value to an applicant predicting their likelihood of default. It has enabled large-scale automation of the application process [85, 86], which is advantageous for lowering costs and for producing more accurate classifications compared to human experts [87]. To assess discrimination, multivariate regression models are a common tool where probability of approval, default, or pricing are dependent variables. One approach in the credit rejection class is exemplified by Han [88], where authors stratified racial disparities according to the length of borrowers' credit histories. To test for taste-based discrimination, a modified regression model was implemented and the coefficient for racial indicator was checked for significant disparities. The study found that the disutility of lending to Blacks persists irrespective of credit history. However, for applicants with no credit history, lenders tended to use observable characteristics (i.e., race/ethnicity) as a proxy for private information, supporting the assumption that lack of data favors statistical discrimination.

²A leading P2P platform based in Beijing, China.

4.2.4 | Lending process stages

Next, it is important to consider where in the lending process discrimination occurs. This can occur at several points from 1) loan advertising and outreach, 2) pre-application enquiries, 3) application considerations to 4) loan approval or denial. We also highlight the category of studies investigating factors external to the direct lending process which can impact any number of these phases. While discrimination may occur at one stage, this can influence the characteristics of potential borrowers in subsequent stages; for example, if a lender systematically encourages minorities to apply for federally insured loans, while encouraging their White counterparts to apply for conventional loans, an analysis becomes problematic since groups are requesting different types of loans regardless of their financial status.

Loan advertising and outreach: Financial exclusion is a major obstacle hindering minorities from entering the marketplace in the first place. Lack of financial literacy, formal employment, and right to property ownership are factors which limit access to and usage of finance [89] with significant disparities observed between genders [90]. In P2P lending, education and university reputation (e.g., graduating from a prestigious university) have a significant effect [71, 91]. Having a child also increased female-headed households' likelihood of being underbanked [92] whereas a couple with children where the female partner participated in the labor market faced familial status discrimination. This effect differed depending on ethnicity [93], exemplifying the contradictory nature of subjective decision making. Some lenders contend that lack of demand explains the relatively fewer mortgages made in minority neighborhoods. Ards et al. [28] found that misinformed self-perception hinders Black applicants from applying in greater numbers. Faced with contentious objections to their creditworthiness, some internalized a negative image to the point that even individuals with a low risk profile believed they had bad credit. This is exacerbated when lenders market loans to minorities and women to a far lesser degree [94], and women are often discouraged from re-applying after a failed attempt [72]. Results also suggest that outcome favorability and opportunity for voice (i.e., the freedom to communicate his/her views and provide information during the decision-making procedures) explain significant variances in customer reactions. Customers responded more positively to unfavorable outcomes from a race-congruent loan officer and female customers responded more positively to low opportunity for voice from male officers than females [95], which ultimately affects loan demand. In summary, the disparity observed further downstream in the lending process already begins to manifest far before the application is even submitted.

Pre-application enquiries: Minorities are less likely to receive information about loan products, they are given less time and encouragement to overcome barriers, and are often quoted higher interest rates compared to their privileged counterparts [61, 96]. Another acknowledged issue are employment practices of lending institutions, particularly in managerial positions. Governance of credit unions by minority directors significantly reduced discrimination effects, particularly in minority neighborhoods where homophily is stronger between directors and applicants [97]. There is anecdotal evidence that racial minorities often feel intimidated when they do not see anyone working "who looks like me" [98]. Lender-lendee homophily partly explains biased decision making. Lenders preferred borrowers with whom they share gender and ethnicity even after controlling for perceived riskiness [60]. In terms of statistical discrimination, the amount of information shared through the credit register is important, particularly between immigrants and natives [80].

Applicant considerations: The use of credit scores can also unfairly disadvantage minorities as they are highly correlated with race/ethnicity, becoming an endogeneity issue although proprietary products [30]. Note that discrimination models assume a single unidirectional causal effect and rarely consider the vicious cycle of discrimination that may have occurred before the target effect. For instance, studies do not typically question whether lower credit scores of minorities are itself a consequence of previous forms of historical injustices.

Credit rejection: Generally, loan approval/denial receives the most interest which can be further organized as credit

rejection, pricing, and default. A disproportionate rate of rejection between groups is considered *prima facie* evidence of discrimination. This strand of literature consists of market-level models, typically by academics, and lender-specific models, typically by bank regulators such as the Office of the Comptroller of the Currency. A study by Blackburn and Vermilyea [44] compared two levels of analysis using HMDA, one at bank-level (i.e., one model per bank) and another at market-level (i.e., one model aggregating variables for several banks), arguing that market-level models produced larger estimated racial effects. Dietrich [45] used ordinary least squares regression to analyze HMDA from 22 banks to unveil multiple discrimination on the grounds of race/ethnicity and gender. In a P2P marketplace, listings with Blacks in attached photos were 25-35% less likely to receive funding compared to their White counterparts [17], undermining P2P's key selling point that investment decisions are collectively made by a crowd thus cancelling individual prediction errors (or biases) [99].

Credit pricing: As competition led to greater variability in loan conditions, a "new inequality" label appeared [48]. Investigations were made into more nuanced terms and conditions, such as whether favorable or subprime [100, 101], average percentage rates [102, 103], interest rates [104, 105, 106, 107], mortgage broker fees [108, 109], contract rates [110, 111] and differences between purchase versus refinancing mortgage loans [39, 112]. Common factors related to pricing differentials include risk elements (credit risk, prepayment risk), expenses (overheads, services), loan characteristics (type and channel of delivery), and negotiation factors [113]. For example, higher priced loans may conceal discrimination if, in addition to the aforementioned factors, race or gender affect the loan price.

Black et al. [114] found that larger overages paid by minorities may reflect discretionary pricing, particularly in negotiation. Applicants often use discretion available to the loan officer to bargain down the rate [113]. Collins et al. [109] found that minority borrowers paid more fees whereas White borrowers more successfully substituted yield spread premiums for up-front cash fees, reducing total fees. Black borrowers were relatively unable to capture the share of decline when rates fell during the rate-lock period [104]. On the other hand, Courchane [103] found that only a small percentage of APR racial differentials could not be explained by typical factors. During the 2000s subprime crisis, studies investigated mortgage redlining in the 'predatory lending' market. Bocian et al. [100] combined HMDA with proprietary subprime loans data and found strong evidence of racial/ethnic segregation. Cheng et al. [105] argued that women pay higher rates because they are more likely to select recommended lenders, whereas men tend to search for the lowest rate. Evidence of a 'glass ceiling' against females in terms of microfinance loan size is also found [81]. We note that contrasting conclusions are frequent largely due to use of various contexts, explanatory variables, and methods.

Credit default: Default-based studies assume that observing different default rates for equally creditworthy groups serves as *prima facie* evidence of discrimination. Berkovec et al. [37] adopted a linear regression model using mortgages insured by the US Federal Housing Administration and found a higher likelihood of default for Black households, justifying higher credit denial. A refined study accounting for market concentration of lenders found greater non-economic discrimination in more concentrated (less competitive) local markets [115]. Anyamele [116] combined annual data from Surveys of Consumer Finances and employed logistic regression and Blinder-Oaxaca decomposition analysis to determine the impact of shocks or triggers to household delinquency rates, finding that 67% of differences between Whites versus Blacks were due to endowments while 33% were unexplained or due to discrimination. They found features such as credit constrained, income, unemployment, payday loan and college graduate to be relevant sources.

Application-external forces: Not all evidence points to the individuals involved in the lending process. Indeed, external factors can also play a significant role in kindling disparity. Institutional characteristics such as its legal structure [117], branch office locations [49], sorting methods [118], risk tolerance [118] and profitability [119] are found to have adverse racial effects. Environmental factors include competition [115], city size [120] and even amount of riot

Phase in credit application	Impact considered	Authors (Year)	Title	Credit Application	Protected characteristic(s) of interest	Data Used	Country
Credit outcome	Traditional vs. ML models	Bono T. (2021)	<i>Algorithmic fairness in credit scoring</i>	General loan	Race / ethnicity; Gender	Credit rating agencies	UK
Credit outcome	Crowd investors vs. ML models	Fu R. (2021)	<i>Crowds, lending, machine, and bias</i>	P2P lending	Race / ethnicity; Gender	Prosper marketplace	US
Credit outcome	Traditional vs. ML models	Tantri P. (2021)	<i>Fintech for the Poor: Financial Intermediation without Discrimination</i>	General loan	Caste	Commercial bank in India	India
Credit outcome	Traditional vs. ML models	Bartlett R. (2022)	<i>Consumer-lending discrimination in the FinTech Era</i>	Mortgage loan	Race / ethnicity	Four mortgage data sources: McDash data ATTOM Data Solutions, HMDA, Equifax	US
Credit outcome	Traditional vs. ML models	Fuster A. (2022)	<i>Predictably Unequal? The Effects of Machine Learning on Credit Markets</i>	Mortgage loan	Race / ethnicity	HMDA with McDashTM mortgage servicing dataset	US
Credit outcome	Traditional vs. ML models	Hauptert (2022)	<i>The Racial Landscape of Fintech Mortgage Lending</i>	Mortgage loan	Race / ethnicity	HMDA	US
Credit outcome	Traditional vs. ML models	Kelley S. (2022)	<i>Antidiscrimination Laws, Artificial Intelligence, and Gender Bias: A Case Study in Nonmortgage Fintech Lending</i>	General loan	Gender	Home Credit (FinTech)	Europe

TABLE 6 Select transitory works comparing traditional and ML approaches.

exposure [121]; for example, riot-exposed Hindu branch managers lend less to Muslim borrowers in relative terms, linking ethnic, intergroup frictions and taste-based discrimination with market efficiency. Other economic factors linked to disparity include inflation in housing prices [122], recession [123], and other market forces established by showing varying levels of discrimination across select time periods. Using a two-stage least-squares regression model, Harvey et al. [124] found that market forces such as improvements in economic conditions and the bank’s financial condition impacted default loss estimates and credit standards in a way that disproportionately benefited minority and low-income applicants. In other words, PC disparities respond to changes in the underwriting standard. These findings underscore the importance of controlling for lender risk assessments when evaluating compliance with fairness statutes. Notably, non-economic factors such as regulatory distinctions play a role as well. Sexual orientation discrimination laws [125], affirmative action policy [22] and degree of conservatism result in high interstate variation, such that gender discrimination is higher in states with low support for gay rights and low female representation in state legislature [126].

Overall, the associated and causal factors attributed to deepening lending inequalities have yet to be fully explored. Contrary to overt prejudice reminiscent of earlier 20th century discriminatory practices have manifested within more clandestine actions and with phantom explanations. We have seen that even well-meaning lenders engage in statistical discrimination, which, regardless of intention, results in the same outcome – limited and unequal access to credit.

4.3 | Transition

As the debate over whether there is discrimination abated (due to undoubtable certainty), developments regarding what steps lenders and regulators should take continued. The financial industry was transformed as a new technological era began with the advent of Big Data and ML. These technologies began to fundamentally transform the way organizations made decisions, resulting in FinTech innovations, operational changes like branchless banking that phases out face-to-face processes, as well as disruptive business models like P2P lending. Two questions stand out as being of particular importance: can state-of-the-art ML models outperform humans and make better (e.g., fairer, more accurate and economical) decisions? Secondly, can machines help remove structural inequalities in society with regards to allocation of financial resources? Researchers begin to question whether machines can help rectify biases or if they risk amplifying them. These are answered via two research streams (Table 6): (1) comparing ML versus traditional credit methods and (2) examining the novel, unique challenges that arise with ML.

4.3.1 | Machine learning vs. traditional methods

Tantri [127] uses application-level data to show ML methods lend up to 60% more at the officers' delinquency rate, improving efficiency without compromising on equity in an emerging market, where soft information traditionally dominates. ML models are observed to be of better predictive quality, more profitable [128], and potentially less discriminatory compared to traditional models like logistic regression. However, this is dependent on PC availability which is often illegal or may require successful PC-proxying. In P2P lending, the "sophisticated" ML algorithm led to a higher rate of return for investors and more opportunities for borrowers as opposed to crowd-based decisions, an improvement even more pronounced for highly risky listings [99].

Some evidence also points to the opposite effect. Certain groups can even 'lose' with more sophisticated technology [129, 130] due to greater model flexibility, which enables algorithms to triangulate the effect of unobserved restricted variables on the outcome by more effectively combining observed permissible variables. This is also reflected in FinTech lending. Whereas rate disparities between White and similar non-White applicants are lower at FinTechs relative to traditional lenders, non-White applicants are still more likely to receive subprime terms for both lender types [131]. This contradicts Bartlett et al. [132], who claimed that FinTechs discriminate less against minorities via interest rates.

4.3.2 | Novel risks of machine learning

Researchers also begin to investigate AI-induced harms to ensure previously seen social and historical biases are not perpetuated into the future. This would further limit resources available to already marginalized groups amidst the claim that FinTechs, for instance, can help 'level the playing field' [133]. Thus, an evaluation of why discrimination is (still) occurring systems, designed to prevent human biases, becomes crucial. And furthermore, whether current regulation provides adequate protection in this new technological environment.

The question faced by researchers at this stage is what drives discrimination? Moreover, what drives the differences in discrimination between statistical and ML models? One explanation is omitted variable bias [134]. The reason is that numerous social, economic, and cultural differences tend to be correlated with racial/ethnic differences, therefore, the omission of any of these features could bias results [135]. In the past, countries began adopting anti-discrimination laws, each encompassing their unique definition of discrimination. However, rapid advancements in AI have outpaced these laws [136], resulting in regulations that may paradoxically hurt rather than help those they are meant to protect. A prominent example is prohibiting the collection of PCs (i.e., 'fairness via unawareness') [60]. A handful of works have investigated the impact of excluding PCs [137, 138, 99]; using U.S. anti-discrimination laws, they concluded that, in the absence of legal constraints, PCs should be used to decrease discrimination and improve predictive quality. Žliobaitė and Custers [139] performed a similar analysis using EU anti-discrimination laws and Andreeva and Matuszyk [140] reached similar conclusions regarding the EU Gender Directive, finding that variable omission led to higher rejection rates for women than men.

4.4 | Optimization

While similar in aim, there is a sizeable shift from focusing on the data to the decision-making process itself with progress in this area largely coming from computer science. We categorize methods for improving fairness based on three key aspects of modelling: (1) selecting input data, (2) defining the outcome constraint and (3) constructing the model. We extend these concepts with our own interpretation of relevant sub-themes.

Purpose	Authors (Year)	Type of analysis or proposal	Fairness criteria assessed	Specific to credit (Y/N)	Additional dimensions	Dataset used	Methodology
Data mining	Berendt B. (2014)	Exploratory discrimination-aware data mining (eDADM)	Differential treatment	No		Case study: Social workers giving advice to a client on loan risk factors	Experiment (Amazon Mechanical Turk)
	Hajian S. (2013)	Direct and indirect discrimination discovery	Direct / Indirect discrimination prevention degree, Direct / Indirect discrimination protection preservation	No		Adult census, German credit	Analytical
	Dietrich J. (2010)	Classification strategy	Marginal effect (coefficient) of subset	Yes		HMDA	Analytical
	Ruggieri S. (2010)	Direct and indirect discrimination discovery: Discriminatory classification rules	Extended lift, α -protective classification rules	No		German credit, PKDD Discovery Challenge 1999 financial dataset	Analytical
	Ruggieri S. (2010)	Discrimination discovery reference model: Logic Programming to Discover Discrimination (LP2DD)	Extended lift, selection lift	No		German credit	Analytical
	Alufaisan Y. (2017)	Discrimination discovery measure: Machine Learning extended lift (MLlift), Feature-based Targeted Sampling (FTS)	Disparate impact	No		Adult census, COMPAS recidivism, German credit	Analytical
Data missingness	Luo L. (2017)	Discrimination-Aware Association Rule classifier (DAAR)	Discrimination score, Inclusion score	No		Adult census, German credit, Traffic Incident Data	Analytical
	Andreeva G. (2019)	Omitted variable bias and proxies	N/A	Yes		Car loans portfolio by major bank	Analytical
	Zhang Y. (2018)	Proxy model: BSG maximum classification	Demographic disparity	Yes		HMDA	Analytical
	Chen J. (2019)	Proxy model: Race proxy based on geolocation	Demographic disparity	Yes		HMDA	Analytical
	Kallus N. (2021)	Proxy model: Partial identification set	Demographic disparity, Negative predictive value disparity, Positive predictive value disparity, True positive rate disparity, True negative rate disparity	No		HMDA, Warfarin dosing PharmGKB	Analytical
	Coston A. (2019)	Weighting methods: Prevalence-constrained covariate shift (PCCS), Target-fair covariate shift (TFCS)	Mean score parity loss, Thresholded score parity loss, Max delta MSP, Max delta TSP	No		Medical expenditure dataset, Mobile money loan approvals in East Africa	Analytical
	Goel N. (2021)	Causal graph-based framework	False positive rate, False negative rate, False discovery rate, Statistical parity, Average odds, Equal opportunity, Disparate impact	No		Adult census, COMPAS recidivism, German credit	Analytical
	Choi Y.J. (2021)	Fair probabilistic circuits (FAIRPC)	Demographic parity	No		Adult census, COMPAS recidivism, German credit	Analytical

TABLE 7 Select studies on dealing with data.

4.4.1 | Dealing with data

What enters the model becomes an area of scrutiny (Table 7), also known as discriminative data mining, in two directions: discrimination discovery and discrimination prevention [141]. These discriminatory contexts, which involve both direct and indirect discrimination³, are modelled using a classification rule-based syntax [142]. Pedreshi et al. [143] were the first to investigate discrimination discovery using an approach based on mining classification rules (inductive) and reasoning on them (deductive) by formalizing legal definitions of discrimination; they proposed an extended lift (elift) measure⁴. Ruggieri et al. [142] extended this framework using the German credit dataset. Alufaisan et al. [141] enhanced the traditional elift measure with mLift given usage of ML for probability estimation. Hajian and Domingo-Ferrer [18] proposed new pre-processing data transformation techniques to remove direct and/or indirect biases with minimal impact. Berendt and Preibusch [144] argued instead for exploratory discrimination-aware data mining (DADM) and asked Mechanical Turk participants, playing the role of a bank clerk or citizen, to recommend or predict approval using discrimination-aware tool support.

Another crucial consideration is data missingness; for example, when PCs or rejected cases are not included in the dataset. HMDA is one of the few regulations in the world which allow lenders to collect information on race/ethnicity and gender of applicants and co-applicants. This is rarely seen in non-mortgage products where self-reported protected basis factors are not available. Limited data availability has motivated the use of probabilistic models to predict and impute the missing PC label based on observable proxies for compliance assessments. Researchers investigate a

³In direct discrimination, the extracted rules can be directly mined in search of discriminatory contexts, whereas in indirect discrimination, more background information is needed (e.g., census data) which, in combination with the extracted rules, allows the unveiling of discriminatory decisions via inference.

⁴Contexts of unlawful discrimination are unveiled by extracting rules, where the degree of burden over protected groups is formalized by an extension of the lift measure of a classification rule. This represents the degree to which the predictions of a classification model are better than randomly generated predictions.

series of proxy approaches using variables like surname and geolocation from Census data and geographic composition (e.g., neighborhood of residence) to impute race/ethnicity [145]. However, these techniques can have opposing, harmful impact – the infamous practice of redlining, for example. When seeing how proxy models fare against fairness outcomes empirically, these methods can even exaggerate disparities [60]. Similarly, Kallus et al. [146] showed that when predicting PCs from proxies, common disparity measures were generally unidentifiable, documenting the biases of these methods.

Because classifiers are trained on historical outcomes, a critical (and often unavoidable) drawback is that they present only one side of reality (i.e., accepted cases only). Another concern are unobservable outcomes or data missingness which, if ignored, can nullify any fairness guarantee of the model once deployed [147]. For example, regarding ‘true’ outcomes, lenders lack information on those originally denied, and whether they would have defaulted had they been approved (i.e., counterfactual); this results in sampling bias or a skewed distribution [148]. In fact, many algorithms cannot guarantee fairness in practice [149]. Using causal graphs, the authors of this study modelled how various past decisions affected data missingness and found that in many scenarios, the distributions used in common fairness algorithms are not recoverable with incomplete data.

4.4.2 | Defining fairness

In an ideal world, a bank would measure discrimination by comparing the probability that a screening model rejects an applicant who will not pay back the loan across PC-defined groups. However, because a bank cannot know the behavior of a rejected applicant, it is forced to use alternative measures. There is still no ‘universal’ definition of fairness due to its complexity; however, over twenty definitions have been proposed [150]. They can be broadly categorized as (1) group fairness [151, 152], (2) individual fairness [153], and (3) counterfactual fairness [154]. For a more exhaustive enumeration, we refer to Dwork et al. [153], Chouldechova and Roth [155].

Most literature focuses on statistical definitions of group fairness [155]. For example, one of the earliest and most common measures of discrimination is raw classification rate or demographic/statistical parity. False positive and negative rates (i.e., equalized odds), which consider the difference of correctly classified instances between groups [151, 156, 157] and positive predictive value [156, 158], the model’s ability to correctly predict outcomes for one group conditional on the known outcome relative to another, also enforce group fairness. These measures are utilized when designing the objective function during model construction [157]. Studies continue to propose new types of definitions (Table 8) such as fairness by explicability, which emphasizes having an external auditor and using explicability via Shapley values⁵ to mitigate bias [160]. Other approaches have proposed non-discrimination norms to adhere to and developed algorithms to verify whether a system violates these norms [161].

To summarize, existing work suggests that (1) measuring a model’s level of fairness is a more complex operation than one might imagine and (2) that results can differ significantly depending on the metric. Fairness notions can also contradict one another [156, 158]. For example, it is impossible to equalize false positive rates, false negative rates, and positive predictive value across groups simultaneously. This is also reflected in inevitable trade-offs between fairness and other desired dimensions or ethical principles, including transparency [162], minority denial rates [163], profit [164] and resource monotonicity [165]. Depending on the bank’s priority, certain axioms are weighted to a greater degree, leading to more (or less) fairness.

⁵Shapley values calculate the average marginal contribution of a select feature towards a model score. This concept is borrowed from cooperative game theory and is a popular ML interpretability tool [159]

Purpose	Authors (Year)	Type of analysis or proposal	Fairness criteria assessed	Specific to credit (Y/N)	Additional dimensions	Dataset used	Methodology
Defining fairness	Goodman L.S. (2019)	Measuring Tool: Real denial rate (RDR)	Observed mortgage denial rate, Real denial rate	Yes		HMDA data with CoreLogic proprietary data	Analytical
	Criado N. (2021)	Normative notions	Explicit direct discrimination, Implicit direct discrimination, Indirect discrimination	No		Adult census, COMPAS recidivism, German credit	Analytical
	Hickey J.M. (2021)	Fairness definition: Fairness by Explicability	Fairness by explicability	No		Adult census, private credit risk dataset, synthetic dataset	Analytical
Fairness trade-offs	Kozodoi N. (2022)	Fairness evaluation	Demographic parity, Equalized odds, Positive predictive value	Yes	Profit	Benelux financial institution, German credit, Give Me Some Credit, Home credit, Taiwan credit, PAKDD, UK financial institution	Analytical
	Lee M.S.A. (2021)	Fairness evaluation	Equal odds, Equal opportunity, False positive error rate balance, Negative class balance, Positive predictive parity, Positive class balance	Yes	Financial inclusion	HMDA mortgage	Analytical
	Gözl P. (2019)	Fairness evaluation	Equalized odds	No	Resource monotonicity, consistency, population monotonicity	FICO score dataset (TransUnion TransRisk scores)	Analytical
	Balashankar A. (2022)	Transparent human intervention	False positive rate, False negative rate, Parity, Pareto losses	Yes	Transparency	Adult census, German credit	Analytical

TABLE 8 Select studies on defining fairness, includes contribution and approach.

4.4.3 | Construction of modelling procedure

The third and most frequently studied aspect of fairness optimization is improving the modelling process itself, for example through regularization, bias mitigation or fair ranking. We analyze how fair modeling methods align with technical notions of fairness (Table 9). Generally, methods that target algorithmic bias can be broadly categorized into pre-, in- or post- processing techniques [166, 20]. This distinction reflects whether they “correct” for fairness at the input level (i.e., training data), in the algorithm (i.e., altering the loss function), or at the output level (i.e., modifying outcomes), respectively.

Studies begin to solve specific problems relevant to the more complicated classification task while forcing it to operate under a particular fairness requirement. For example, there is the overall classification error problem with class imbalance, where model performance for the minority class is often ignored to give priority to the majority, resulting in low true positive rates (TPR) and high true negative rates (TNR). An approach known as AdaFair attempts to overcome this by preserving high TPRs and TNRs for both the discriminated-against and privileged group [167]. Another barrier of lending to higher risk, underserved populations is the time consuming and expensive screening process. Taking the perspective of a budget-constrained decision-maker, Cai et al. [168] derived an algorithm using a utility maximizing strategy to solve the screening and allocation problem optimally. Researchers also look to specific models, like graph neural networks with a NIFTY (uNifying Fairness and stabiliTY) framework to learn fair and stable representations [169] and using deep networks via continual learning [170].

To address the impossibility theorem,⁶ researchers look to combine multiple fairness constraints in a single framework [171]. Martinez et al. [172] showed how fairness can be formulated as a multi-objective optimization problem and used Pareto optimality to select the most efficient classifier with the smallest worst group conditional risk. Researchers also try ensuring fairness alongside additional dimensions like explainability and interpretability [173, 2], which are crucial to trustworthy design. Explainable AI approaches are shown to improve fairness [170], and template workflows are proposed which combine interpretable models, post-hoc explanations, and discrimination testing, all in one software tool [173].

⁶Multiple fairness definitions cannot be satisfied simultaneously.

Purpose	Authors (Year)	Type of analysis or proposal	Fairness criteria assessed	Specific to credit (Y/N)	Additional dimensions	Dataset used	Methodology
Bias mitigation	Chakraborty J. (2021)	Pre-processing for data imbalance: Fair-SMOTE	Average odds, Equal opportunity, Statistical parity, Disparate impact	No		Adult census, Bank marketing, COMPAS recidivism, Default credit, German credit, Heart health, Home credit, MEPS-15,16, Student performance	Analytical
	Singh (2022)	Bias mitigation technique: DualFair	Alternate world index, Average odds difference	No		HMDA mortgage	Analytical
Fair ranking	Cai W. (2020)	Candidate screening and resource allocation optimization	Group-specific utilities	No		German credit, synthetic dataset	Analytical
	Schoeffer J. (2021)	Fair ranking: based classification algorithm	Fairness through awareness, Fairness through unawareness	No		German credit, synthetic dataset	Analytical
	Liu W. (2019)	Fairness-aware re-ranking: Personalized Fairness-Aware Re-ranking (PFAR)	Average coverage rate	Yes		Kiva microloans	Analytical
Enhancing fairness	Lee E.L. (2014)	Fairness-aware classifier: Bayesian Personalized Ranking optimization criterion coupled with matrix factorization (BPRMF)	Standard deviation of recommendation frequency to lender	Yes		Kiva (micro-loan platform)	Analytical
	Agarwal C. (2021)	Fairness-aware classifier: unifying Fairness and stability (UNIFY)	Counterfactual fairness, Equal opportunity, Statistical parity	No	Stability	COMPAS recidivism, Credit defaulter, German credit	Analytical
	Iosifidis V. (2019)	Fairness-aware classifier: Adafair	Equalized odds	No		Adult census, Bank marketing, COMPAS recidivism, KDD census income	Analytical
	Reich C.L. (2021)	Fairness-aware classifier: calibration and equal error rate convergence	Calibration, Equal error rates	No		COMPAS recidivism, Survey of Income and Program Participation dataset	Analytical
	Sun Y. (2021)	Fairness-aware classifier: G- FORCE (Group-Fair, Optimal, Randomized Combination of Experts)	Equalized odds	No		Adult census, German credit and COMPAS recidivism	Analytical
	Wagner B. (2021)	Fairness-aware classifier: Neural-symbolic approaches	Demographic parity, Disparate impact	No	Explainability	Adult census, COMPAS recidivism, German credit	Analytical
	Martinez N. (2020)	Fairness-aware classifier: minimax Pareto fairness (MMPF)	Group fairness in terms of predictive risk disparities	No		Adult census, German credit, ICU mortality rates, Skin lesions	Analytical
	Kanamori K. (2021)	Post-processing framework: Fairness-Aware Decision tree Editing (FADE)	Demographic parity	No		German credit, COMPAS recidivism	Analytical
	Khalili M.M. (2021)	Post-processing framework: Improving privacy and fairness	Equal opportunity	No	Privacy	FICO score dataset (TransUnion TransRisk scores), Synthetic dataset	Analytical
	Gill N. (2020)	ML template workflow	Adverse impact ratio, Marginal effect, Standardized mean difference	No	Interpretability, Explainability	HMDA mortgage	Analytical
	Castelnovo A. (2021)	Fairness-aware classifier: Individual and group fairness convergence	Demographic Parity	Yes	Explainability	Intesa Sanpaolo bank loans	Analytical

TABLE 9 Select fairness framework studies.

4.5 | Performance

4.5.1 | Benchmarking

Exploratory works that perform broad analyses and benchmarking of fair modelling techniques (Table 10) quickly followed the development of the above techniques. For example, bias mitigation techniques, including pre-processing, in-processing and post-processing*Pre-processing methodologies include* re-weighting *and* disparate impact removal, *in-processing includes* adversarial debiasing *and* prejudice remover regularization *and post-processing includes* equalized odds, calibrated equalized odds, *and* reject option classification. are explored by Biswas and Rajan [174] and Stevens et al. [175]. Islam et al. [176] extended this by evaluating 13 fair classification approaches in terms of their correctness, fairness, efficiency, scalability, robustness, sensitivity, data efficiency, and stability. These were measured along multiple fairness metrics, datasets, and classifier types [177]. Li et al. [178] and Stevens et al. [175] are the only two works in this category to apply testing solely on credit-related datasets (Kiva and Prosper), whereas others assess fairness across multiple domains including income, recidivism, product subscription and passenger survival.

4.5.2 | Tools

With growing interest in “auditing” AI, public and private institutions have launched user-friendly support toolkits to prevent unjust discrimination prior to deployment [179]. Major efforts have been made by IBM with AIF360 [180], Microsoft with Fairlearn [181], Google with What-If tool [182] and University of Chicago with Aequitas [183]. For example, AIF360 incorporates 70+ metrics and algorithms to detect bias in datasets and models. Others have introduced packages like Fairmodels to help visualize and eliminate bias [184] and NeuronFair, which is tailored to deep neural networks and enhanced by interpretability and generalizability to handle structured and unstructured

Purpose	Authors (Year)	Type of analysis or proposal	Fairness criteria assessed	Specific to credit (Y/N)	Additional dimensions	Dataset used	Methodology
Fairness testing	Islam M.T. (2022)	Evaluation & Benchmarking	Disparate impact, Individual discrimination, Total effect, True negative rate balance, True positive rate balance	No		Adult census, COMPAS recidivism, German credit	Analytical
	Biswas S. (2020)	Evaluation & Benchmarking	Average odds difference, Consistency, Disparate impact, Error rate difference, Equal opportunity difference, Statistical parity difference, Theil index	No		Adult census, Bank marketing, German credit, Home credit, Titanic	Analytical
	Teodorescu M.H.M. (2021)	Evaluation & Benchmarking	Demographic parity, Equalized opportunity, Equalized odds, Fairness through unawareness	No		Bank Marketing	Analytical
	Stevens A. (2020)	Evaluation & Benchmarking	Average absolute odds, Disparate impact, Equal opportunity, Statistical parity	Yes	Explainability	Kiva loans	Analytical
	Li Y. (2020)	Evaluation & Benchmarking	Counterfactual fairness, Individual fairness	Yes	Social scoring	Prosper loans social network	Analytical
Fairness tool	Zheng H. (2022)	Fairness toolkit: NeuronFair	Individual discriminatory instances	No	Interpretability	Adult census, Bank marketing, COMPAS recidivism, German credit, Medical expenditure, Image datasets	Analytical
	Bellamy R.K.E. (2019)	Fairness toolkit: AIF360	Average odds difference, Disparate impact, Equal opportunity difference, Statistical parity difference	No		Adult Census, COMPAS recidivism, German Credit	Analytical
	Wiśniewski J. (2022)	Fairness toolkit: fairmodels	Equal opportunity, Overall accuracy equality, Predictive parity, Predictive equality, Statistical	No		German credit	Analytical
	Bantilan N. (2018)	User interface design: Themis-ml	Group-level discrimination (mean difference, normalized mean difference), Individual-level discrimination (consistency, situation test score)	No		German credit	Analytical
	Nakao (2022)	User interface design: FairHIL	Average odds difference, Disparate impact, Equality of opportunity difference, Statistical parity difference, Theil index	Yes		Anonymous bank loan dataset	Think-aloud user study (w/ loan officers & data
	Purificato (2022)	User interface design	Disparate impact	Yes	Explainability	Case study: Bank loan decisions	Experiment (w/ loan officers and end-users)
	Nakao Y. (2022)	User interface design	Equality of opportunity, Individual fairness	Yes	Interpretability	Case study: Bank loan decisions	Experiment (w/ end-users)

TABLE 10 Sample of fairness performance studies.

datasets [21]. Many studies validate their methodology on multiple benchmarking datasets as they aim to provide generic tools for any domain.

More recently, researchers have been involving end-users and those without any technical background in a human-in-the-loop process [185]. User interfaces are designed to support external stakeholders investigating fairness [186, 185] while considering principles like explainability and interpretability [187]. One stark but positive feature of the latter two works is the involvement of loan officers and data scientists directly in the testing process. Loan officers were able to leverage their domain knowledge, resulting in the proposal for ‘custom’ attributes to support fairness assessments. In the case of financial services, capturing the notion of *affordability* is a key concern.

4.6 | Application

4.6.1 | Long-term fairness

Much prior work focuses on decisions made in static, or one-shot, decision contexts. The long-term impact on the population, feedback loops as well as system-level dynamics are rarely considered and remain relatively under-studied (Table 11). The lasting effect of a decision made by a learning agent can conflict with fairness objectives if optimized in a static setting [188]. In practice, decisions can affect feature distributions across sub-populations and thus influence future rewards. This can eventually affect the dynamics of a decision-making loop. For example, consider that successful applicants from a particular group may inspire or recommend similar candidates noting positive reviews, thus affecting subsequent applicant pools. This strategy enables banks to learn more about this subpopulation, which may encourage richer pools of qualified applicants going forward (i.e., positive reinforcement). Puranik et al. [189] modelled these long-term dynamic effects using a reinforcement learning framework based on Markov Decision Processes (MDP). They assigned two objectives – maximizing utility by admitting candidates with the highest scores and (2) minimizing disparity between the proportions approved from each group and proposed a ‘Fair-Greedy’ policy.

Correspondingly, D’Amour et al. [190] showed that simulation studies can serve as a helpful testbed for systematically exploring long-term implications. They enable access to counterfactual information, which shows how data would have varied if a different collection or decision-making policy was made – a missing dimension in static datasets.

Purpose	Authors (Year)	Type of analysis or proposal	Fairness criteria assessed	Specific to credit (Y/N)	Additional dimensions	Dataset used	Methodology
Long-term fairness							
	Creager E. (2020)	Causal framework: Causal directed acyclic graphs (DAGs)	Demographic parity, Equal opportunity	Yes		FICO dataset (TransUnion TransRisk scores)	Analytical
	Puranik B. (2022)	Fair selection in sequential decision making: Fair-Greedy policy	Demographic parity	No		Law school bar study, German credit	Analytical
	D'Amour A. (2020)	Simulation toolkit: ml-fairness-gym	Equality of opportunity, Equality of discovery probability	No		FICO score dataset (TransUnion TransRisk scores)	Analytical
	Wen (2021)	Fair policies which account for feedback effects	Demographic parity, Equal opportunity	No		FICO score dataset (TransUnion TransRisk scores)	Analytical
Perception of fairness							
	Albach M. (2021)	Fairness of the process	Perception of AI fairness on a 7-point Likert scale	No		Child protective services, COMPAS recidivism, Hospital resources, Insurance rates, Loans, Unemployment aid	Survey (Amazon Mechanical Turk workers)
	Gemalmaz M.A. (2022)	Fairness of the outcome	Subject's perceived fairness level, Subject's retention in the AI system, 5-point Likert scale	Yes		Bank loan decisions	Human-subject experiments
	Saxena N.A. (2020)	Fairness of the outcome	Treating similar individuals similarly, Never favor a worse individual over a better one, Calibrated fairness	Yes		Bank loan decisions	Survey (Amazon Mechanical Turk workers)
	Schoeffer J. (2022)	Fairness of the process (information received)	Agreement along 5-point Likert scale; Open-ended responses	Yes	Trustworthiness	Bank loan decisions	Human-subject experiments
	Worthington S. (2013)	Fairness of the process (treatment by financial services provider)	Fairness constructs measured in the survey: Procedural fairness, Interactional fairness, Distributive fairness	Yes		Bank loan decisions	Survey (customers)

TABLE 11 Select fairness application studies.

One example is a dynamic lending scenario where an applicant's credit score can change in response to an agent's decision to approve/reject. Wen et al. [191] simulated a bank lending scenario by modeling feedback effects as MDPs. They distinguished outcome quality for the decision maker (bank) from outcome quality for the applicant and imposed fairness properties on the average outcome quality for individuals across subgroups, as well as reward measures for the decision maker (profit). By understanding how learning agents interact with ML systems in a simulation, these works incorporate more context and highlight that static, single-step analyses do not provide a comprehensive picture of long-term consequences.

4.6.2 | User Perception

Until now, the discussed works have often pursued fairness by defining it in technical terms. However, some consider this a one-sided 'techno-centric' view which ignores the attitudes and opinions of those about whom decisions are being made. Consequently, a relatively new stream of research in AI and human-computer interaction emerged that treats fairness as a subjective concept and examines users' perceptions of what "fair" means and, correspondingly, what factors affect this perception. To this end, Saxena et al. [192] compared three fairness definitions and investigated how lay persons perceived them in a lending context. They showed preference for *calibrated fairness* and public support for affirmative action, where revealing sensitive PC information influenced respondents' perception. Gemalmaz and Yin [193] used randomized human subject experiments and found subjects were significantly affected by whether the system made decisions in favor of the group the subject belongs to, as opposed to unbiased treatment. Another line of research examines moral reasoning. Albach and Wright [194] analyzed different scenarios from bail and hospital resource allocation to lending, and asked survey respondents to judge how fair it is to use certain features in modelling. In line with safeguarding responsible AI principles such as transparency and accountability, like the "right to explanation" in the GDPR, Schoeffer et al. [195] examined the effect of an explanation on subjects' perceptions regarding informational fairness (i.e., whether they were provided adequate information and explanation) and overall trustworthiness.

5 | DISCUSSION AND CONCLUSION

Discrimination and fairness in credit have attracted significant scholarly interest over the last 40 years. However, the body of knowledge produced is fragmented and dispersed across multiple disciplines (e.g., economics, business and management, social sciences, ethics, computer science), and lacks a systematic view into their conceptual connections. Therefore, the purpose of this study is to critically review and systematize the extant literature and scrutinize the growing role of technology (i.e., ML) as a key factor that continues to change the field's trajectory. We systematize this in the framework (Figure 5) covering different lending process stages, its relationship to technology and the real-world environment (i.e., policy, social context, business market), and how they form the causes, consequents, and solutions for achieving fair outcomes.

We observe that most work looks at distinct segments, solely from a technological, economic or sociological point of view, however, this inhibits cross-disciplinary insights. For example, some reviews address fairness in ML but lack the credit context [155, 150, 196], or explored discrimination in credit systems but lack ML consideration [59, 197, 198, 199], while others have examined ML in credit systems however lack an understanding of fairness, focusing rather on technology enhancement of predictive capability [200, 201, 202, 5] (Figure 7). A gap remains in terms of linking the genesis and outcomes that underpin these distinct streams. To the best of our knowledge, this study is the first to review all segments of discrimination and fairness work in this context.

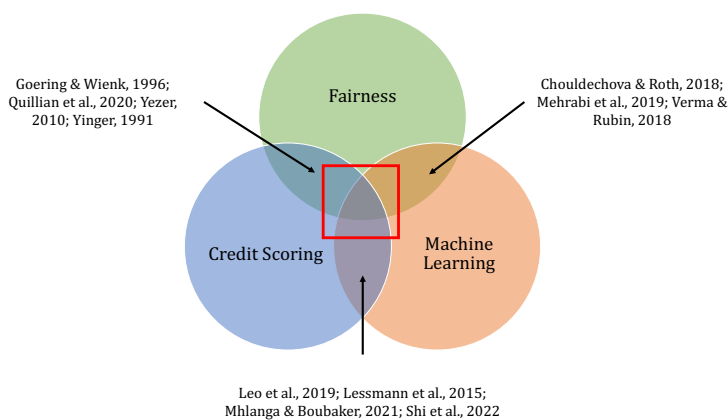


FIGURE 7 Three topics of interest and their overlaps with relevant literature reviews. Note: Red box highlights the gap.

5.1 | Research Gaps and Future Directions

5.1.1 | Credit focus

It is clear from profiling that an AI-inspired stream of research is in the making and we are likely to observe a progressive flow of studies on how ML can help achieve fairness objectives in society. However, a major gap is the lack of a credit-specific focus in this line of work. Data analysts often view fairness from a generalizable, application-agnostic perspective, evidenced by the high number of studies that utilize multiple, diverse datasets to validate their approach. The focus is not credit and its nuances, but an investigation into an algorithm or data processing technique which

can be applied ubiquitously. An associated consequence is an overall detachment from the real-world application of technologies in a specific sector, resulting in lack of engagement with the business community, e.g., to help inform or assess the data-driven methodologies of individual organizations, alongside data sharing opportunities.

We also highlight the oddly minimal number of publications on the relationship between digital transformation and its consequential risks in well-respected sources, particularly those specializing in business and management. This is in stark contrast to the abundant attention paid to ML for enhancing predictability of credit models [203, 202, 204]. A small number of works have started exploring ML with explainability and interpretability goals [205, 206], however, there has not yet been a similar upsurge for fairness. From this viewpoint, the management literature fails to fulfill its role of guiding companies in implementing proper strategies and updating regulation. It is crucial that the topic not only be considered by IT journals, as it currently is, but by the widest possible community of scholars in its connection to financial markets.

5.1.2 | Data limitations

The second gap relates to data availability, particularly lack of data and that of *diverse* data. Legal constraints prohibit PC collection; however, countless studies have highlighted the consequences of omitted variable bias, including suboptimal proxies, imputation flaws and hindrances to auditing. This brings us to the second issue – lack of data diversity, as evidenced by the oversaturated usage of the outdated German credit dataset from 1994. Taking precedent from HMDA, which has historically been effective due to its validation potential, similar availability of PC information would be beneficial. Studies are recommended to make the data used in their analysis public in order to populate the open-source domain; however, we acknowledge the difficulty given the proprietary nature of financial services.

5.1.3 | Intersectionality

We note the minimal attention paid to intersectionality, a term coined by legal scholar Kimberlé Crenshaw to express the multiple threats of discrimination which might occur when an individual's identities overlap and intersect [207]. Individuals often belong to multiple categories simultaneously, across gender, sexual orientation, race, religion, or national origin (e.g., homosexual white female). We highlight the limited number of works which study multiple PCs simultaneously as well as multinominal characteristics and their combinatorial effects, thereby disregarding the multifaceted nature of human experience and social position. For example, the German credit and Adult census datasets are assessed for gender, while COMPAS only considers ethnicity [176]. More intersectionality-cognizant work would help refocus on the practical aspects of fairness solutions.

5.1.4 | Strategic studies

Overarching strategic studies are scarce. Our review reveals a limited number of works looking at the overall workflow and how the variety of proposals interact across the entire lending process. Among the few exceptions are Gill et al. [173] on ML workflow as well as the emerging work on dynamical systems which considers long-term impact [208, 189]. Holistic solutions are crucial, as they view fairness as a 'process' and not solvable via one-shot solutions which only cover individual lending phases. This would significantly contribute to addressing the transferability issue and ensuring practicality of findings.

5.1.5 | Application-level studies

The final gap is the lack of studies conducted at the organizational or application level. It was noted that most empirical works focus on the theoretical and experimental nature of fairness validated using toy datasets. The lack of knowledge on whether and how financial institutions themselves pursue fairness objectives in this AI age is currently one of the main impediments of research, regulatory catch-up as well as lack of institutional standards [209]. In other words, there is a theory-to-practice gap – a limited understanding of how tools proposed in academia are (or are not) being applied in the real world. Questions that need answering include: What are banks doing to ensure fair outcomes? What thresholds are they considering as fair (*enough*)? Are consistent methodologies being implemented which can be benchmarked across multiple banks? While this insider information is of high interest to banks individually, lack of public disclosure by these same banks, likely due to risk of exposure and “first-mover disadvantage,” results in continued ambiguity.

5.2 | Contributions and Implications

Our findings offer significant contributions to researchers from several disciplines. Firstly, this study pioneers research on fair credit from a statistical, computational, and social lens with a unifying objective. We have identified six thematic foci, finding that the advancement of the topic is subject to complexities at the nexus of several lending phases and contextual considerations which interact at various levels. Specifically, our framework defines the boundaries of discrimination-causing, -mitigating and -consequential elements across the lending process. It identifies the spectrum of factors to consider regarding data and modeling which can help overcome weaknesses posed at any other lending phase; this can help streamline the development of automated or semi-automated workflows. Thus, developing the topic as advocated in this study directly benefits several stakeholders, including financial institutions, managers and heads of functions, product developers, software engineers, public administrators and regulators looking to safeguard fair lending objectives.

In tandem, the lack of claimed problem ownership to a specific discipline is observed. Because the concept of fairness is loaded with historical underpinnings, it is subject to individual as well as societal judgement, and manifests as a (partially) solvable technical issue, thereby inviting multiple communities to take up interest. Thus, our study is a comprehensive, multi-disciplinary analysis of the status quo, which serves as basis for the development of new research connecting different macro-topics more aligned with the practice environment. By highlighting the missing themes that bridge the disjointed works as well as the odd under-consideration from highly ranked journals, we hope this study serves to re-establish research equilibria.

Moreover, a major issue encountered in the extant literature is that the available information is often too general and context-agnostic, thus not easily transferable to practice. Additional requirements are needed for application, such as standardized norms around minimum disparity thresholds and a clearly defined definition for fairness applicable to credit-related outcomes. For instance, the cost of misclassification differs for incarcerating an individual or firing an employee as opposed to product recommendations or online translations which have milder consequences [210]. Additionally, whether the decision results in an opportunity (i.e., benefit) or punishment (i.e., harm) plays a major role; for example, in criminal risk assessment, it is commonly accepted that incarcerating an innocent person is more serious than letting a guilty person escape (i.e., false negative). It is up to the financial services industry to define these preferences. We highlight Kozodoi et al. [164] as one of the few studies to claim *separation* is the most appropriate metric for credit scoring.

Finally, poor management of biased outcomes can lead to negative customer sentiments, press, and regulatory

finances [163], which can ultimately impact a company's reputation and financial performance. Thus, managers need to understand the many interacting components before an applicant has even walked through the door, as well as the long-term effects across multiple applicant cycles. By considering fairness across the entire modelling pipeline, financial institutions can also ensure their survival by way of compliance with more stringent regulations on the horizon; for example, the Artificial Intelligence Act is an upcoming European regulation which will regulate AI across all sectors. Additionally, companies can engage with stakeholders and regulators to help set industry standards. Take, for example, the airline industry: in parallel with setting-up a single aviation market, common rules have been adopted to ensure its proper functioning which require, notably, (1) a level playing field, or in our case, equal levels of access to data and PCs for auditing, and (2) a high and uniform level of protection for passengers, or in our case, unbiased outcomes for customers. Similarly, the industry must codify a set of common reporting standards so banks can adjust and invest in areas where checks can be most effective, and regulators can determine which interventions show the greatest benefit to set comparable standards. Responsible technology use can also serve as an incentive for acquiring customers who want to engage with 'trustworthy' businesses, like customers 'going green' making shopping decisions based on companies' sustainability principles. In sum, managers need to acknowledge the importance of fair decision making and realize there are no obvious answers. However, taking steps towards meaningful change can help firms stay consistent, up-to-date, and trustworthy in the AI-driven environment.

In this article, we reviewed the 40-year evolution of research in fairness and discrimination in credit, reducing the ambiguity that plagues the field characterized by causes, consequences and solutions spanning cross-disciplinary works. These insights can help readers understand the current state-of-the-art and its materialization, although fragmented. We hope to inspire holistic, yet application-cognizant solutions going forward to help disentangle the complexity of fairness.

Appendix

Distribution of articles by journal and discipline

Journal/Conference title	1978-1990	1991-2000	2000-2010	2011-2022	Total articles	%
Journal of Financial Services Research	0	3	2	4	9	4.9
Housing Policy Debate	0	0	1	5	6	3.2
Journal of Real Estate Finance And Economics	0	1	1	4	6	3.2
Review of Black Political Economy	0	0	2	2	4	2.2
AAAI Conference on Artificial Intelligence Proceedings	0	0	0	3	3	1.6
ACM International Conference Proceeding	0	0	0	3	3	1.6
Journal of Banking and Finance	0	0	1	2	3	1.6
Journal of Urban Economics	0	0	2	1	3	1.6
Lecture Notes in Computer Science	0	0	0	3	3	1.6
Review of Financial Studies	0	0	0	3	3	1.6
Social Science Quarterly	0	3	0	0	3	1.6
World Development	0	0	0	3	3	1.6
International Conference on Machine Learning (ICML) Proceedings	0	0	0	2	2	1.1
AAAI/ACM Conference on AI, Ethics, and Society (AIES) Proceedings	0	0	0	2	2	1.1
American Economic Review	1	1	0	0	2	1.1
Other journals (one or less)	5	11	12	102	130	70.3
Total No. of Articles	6	19	21	139	185	100.0
%	3.2	10.3	11.4	75.1	100.0	

TABLE A.1 Top journals and conferences over time, across four balanced time periods.

Discipline	1978-1990	1991-2000	2000-2010	2011-2022	Total articles	%
AI Ethics	0	0	0	6	6	3.2
Business, Finance, Economics	5	6	11	53	75	40.5
Computer Science	0	0	2	42	44	23.8
Development Studies	0	0	0	5	5	2.7
Human Computer Interaction	0	0	0	3	3	1.6
Mathematics and Statistics	0	0	0	2	2	1.1
Social Sciences	1	13	8	28	50	27.0
Total No. of Articles	6	19	21	139	185	100.0
%	3.2	10.3	11.4	75.1	100.0	

TABLE A.2 Disciplines included in the final sample over time.

Geographic analysis of authorship origin

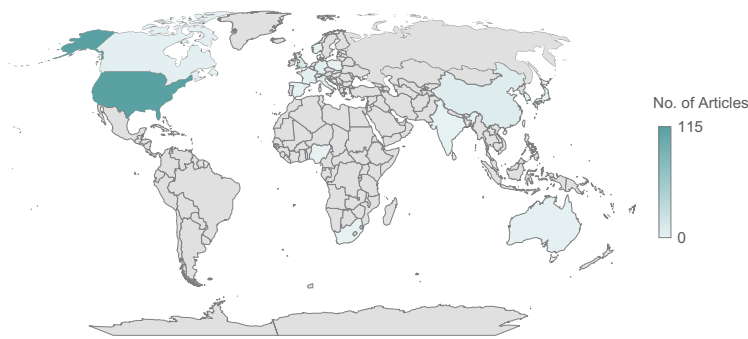


FIGURE A.1 Spatial representation of authorship origin. *Note: Calculated based on the affiliation of the first author of each publication; in the case the author has several affiliations, the first-mentioned affiliation is selected.*

Credit application coverage

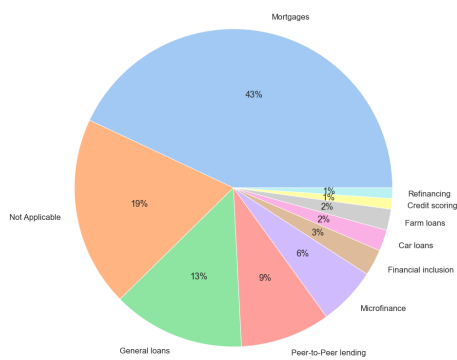


FIGURE A.2 Distribution of credit application contexts studied.

Keywords co-occurrence

With a threshold of five recurrences, we identify 31 most commonly co-occurring terminologies out of 923 keywords. As per our expectations, ‘discrimination,’ which forms our conceptual groundwork, emerges as the most co-cited phrase (n=36). This is accompanied by ‘machine learning’ (n=27), ‘decision making’ (n=24), ‘mortgage lending’ (n=22), ‘race’ (n=12), ‘gender’ (n=12), ‘classification of information’ (n=11), ‘fairness’ (n=11), ‘microfinance’ (n=11). We note that ML becomes a dominant concept towards 2020 (yellow), particularly learning systems, data mining and FinTechs. Consequently, the transforming data environment triggers concern over social and ethical issues within ADM systems.

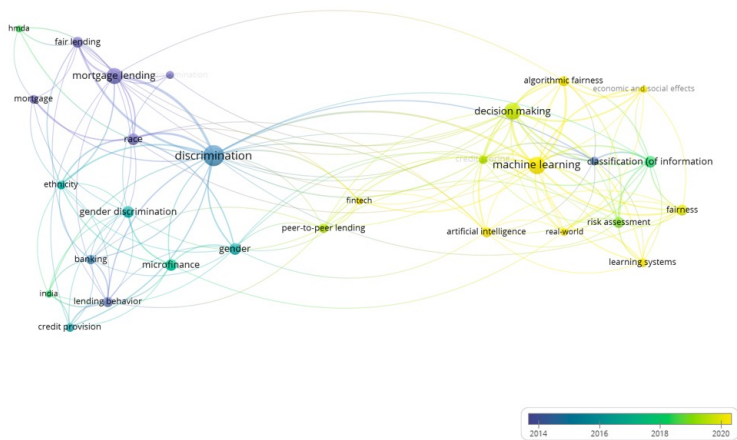


FIGURE A.3 Dynamic keywords co-occurrence analysis using VOSviewer.

Authors (Year)	Credit Application	Protected characteristic(s) of interest	Data used	Country	Unwarranted discrimination	Key Findings
Black B.H. (1978)	Mortgage loan	Race / ethnicity	Survey instruments mailed to 300 institutions	US	Y	Race is an important determinant in the loan decision at the 90% level of confidence; blacks are less likely to be granted loans than non-blacks, ceteris paribus.
Peterson R.L. (1981)	General loan	Gender	Federal Reserve survey	US	N	No systematic pattern of preludinal sex discrimination was found, even before the EDA was passed. Banks behaved as profit maximizers, making loans on equivalent terms to equally risky customers, regardless of sex.
Ingram F.J. (1981)	Mortgage loan	Race / ethnicity; Gender; Age	Four largest commercial banks and all four savings and loan associations	US	N	Protected classes of applicants such as minorities, women, and the elderly were afforded the same access to credit as their unprotected counterparts in the same risk categories.
Guy R.E. (1982)	Mortgage loan	Race / ethnicity	HMDA	US	Y	Beyond instituting objective criteria (i.e., income, neighborhood conditions, and two dimensions of demand), the percent of the population with black significantly affects (negatively) lending rates.
Leahy P.J. (1985)	Mortgage loan	Race / ethnicity	Conventional mortgage loans by 12 largest lenders in Summit County, Ohio	US	Y	Conventional mortgage loans by number of mortgage loans and amount of money loaned; black tracts receive significantly less of each.
Shaw W.B. (1985)	Mortgage loan	Race / ethnicity	Annual Housing Survey SACS samples	US	Y	Younger households participate more heavily in the FHA sector than other households; racial discrimination does occur in the conventional mortgage-lending sector and with consequent heavy participation in the FHA.
Schill M.H. (1994)	Mortgage loan	Race / ethnicity	HMDA and the Census	US	Y	In predominantly black neighborhoods, the rejection rates for black loan applicants are actually lower than those for white loan applicants.
Myers S.L. (1995)	Mortgage loan	Race / ethnicity	HMDA	US	Y	70% of the racial gap in loan denial rates in New Jersey can be accounted for by racially discriminatory lending practices; differences in bad credit FHA account for only 9% of the nonindemnity portion of the racial gap in lending.
Mumml A.H. (1996)	Mortgage loan	Race / ethnicity	HMDA, Boston lending institutions	US	Y	Minorities are more than twice as likely to be denied a mortgage as whites.
Squires G.D. (1996)	Mortgage loan	Race / ethnicity	HMDA	US	Y	Black applicants and those from predominantly black communities are less likely to have their loan applications approved than are white applicants or applicants from predominantly white areas.
Rosenblatt E. (1997)	Mortgage loan	Race / ethnicity	New Jersey-based City Federal Savings Bank	US	Y	Contrary to common assumptions, race differences in FHA VA Indirect are at least as pronounced as in conventional lending, and outcomes for Asians, diverge as much from outcomes for whites, as do outcomes for Hispanics and African American.
Gotham K.E. (1998)	Mortgage loan	Race / ethnicity	HMDA	US	Y	High-income African Americans are rejected at higher rates than low-income whites; the low percentage of minority applications suggests meager marketing and pre-screening where they are discouraged from completing an application form.
Holloway S.R. (1998)	Mortgage loan	Race / ethnicity	HMDA	US	Y	The effect of being black on the conditional probability of having a loan application denied is much higher in all-white neighborhoods than in minority neighborhoods, especially for applicants making large loan requests.
Ehly D.R. (1999)	Mortgage loan	Age	National Association of Realtors' data set	US	Y	Differences between the residential mortgage loan bundles of the two groups by age of the applicant were revealed.
Martin R.E. (2000)	Car loan	Race / ethnicity	Single lender who makes loans in the "Car and Used car market"	US	Y	Administration costs and default rates are higher for minorities than for whites, controlling for age, income, home ownership, wealth, occupation, loan terms, and geographic location.
Reibel M. (2000)	Mortgage loan	Race / ethnicity	HMDA	US	Y	Models for mortgage application approval show across-the-board discrimination against Black applicants but not against Hispanic applicants; applicants in both Black and Hispanic neighborhoods, however, are systematically denied loans.
McMurray J.P. (2003)	Mortgage loan	Race / ethnicity	Nationally-oriented mortgage originator	US	N	No statistical difference found between minority and non-minority applicants, indicating that originators do not demonstrate a taste for discrimination but take their loan approval for minorities on whether the loan can be profitably sold.
Greer K.E. (2004)	Mortgage loan	Race / ethnicity	HMDA; Bank data from Call and Income reports, Census	US	Y	Native American applicants experience higher conditional denial rates compared with similar White applicants, suggesting lack of financial resources, applicant education about and experience with mortgage processes, low creditworthiness, and lender reluctance to confront but disburse the necessary limit on reservation guarantee program success.
Gary-Bobo R.L. (2004)	Mortgage loan	Occupation	French mortgage lender, the Credit Hypothecaire de France	France	Y	Presence of "social discrimination" in the data where the loan conditions depend not only on the borrower's wage and down payment, but also on the borrower's occupational status.
Blank E.C. (2005)	Mortgage loan	Race / ethnicity	HMDA	US	Y	Blacks have a lower probability of having their applications approved than whites, similar for Hispanics; reveals several vulnerable points in the loan approval process where individual or institutional biases could occur, including disproportionately low number of minority loan officers, statistical bias against minority applicants, and bias built into the Fair Isaac credit scoring system.
Bodinn T.P. (2006)	Mortgage loan	Race / ethnicity	American Housing Survey	US	Y	Significant differences are more likely in the conventional mortgage market with largest among blacks, who pay a higher APR than whites for both purchases and refinancing; for government-insured loans, Hispanics do slightly better than whites.
Courchane M.J. (2007)	Mortgage loan	Race / ethnicity	HMDA	US	N	50% of the African American APR gap, and 85% of the Hispanic APR gap is attributable to observable differences in underwriting, costing, and market factors that appropriately explain mortgage pricing differentials.
Hodge M.E. (2007)	Refinancing	Race / ethnicity	Data from southeastern regional bank	US	Y	African Americans are significantly more likely to be turned down for mortgage refinancing than their white counterparts; refinancing discrimination is one contributing factor to the continuing wealth or net worth gaps.
Bocian D.G. (2008)	Mortgage loan	Race / ethnicity	HMDA; proprietary database of securitized subprime loans	US	Y	African American and Latino borrowers are more likely to receive higher-rate subprime home loans than non-Latino white borrowers.
Ezalde-Harrison F. (2008)	Mortgage loan	Race / ethnicity	HMDA; Western Economic Services	US	Y	Consistently high denial rates for minorities, which is not explained by financial (economic) factors.
Ezalde-Harrison F. (2008)	Mortgage loan	Race / ethnicity	HMDA; Department of Housing and Urban Development data	US	Y	Consistently high denial rates against minorities indicating strong evidence of redlining practices in housing loan decisions thus contributing to the growth of racial segregation in the state.
Ross S.L. (2008)	Mortgage loan	Race / ethnicity	Mortgage lending institutions in Los Angeles, California and Chicago, Illinois	US	Y	In Chicago, African American and Hispanics systematically received less information and assistance than comparable whites; while in Los Angeles, the treatment of minority and white testers did not differ statistically from each other.
Escalante C.L. (2009)	Farm loan	Gender	Loan application database of the Georgia FSA State office	US	N	Results do not produce overwhelming evidence of gender bias in FSA loan approval decisions and in favor of the male gender.
Hirasawa (2012)	Mortgage loan	Race / ethnicity; Marriage status / sexual orientation	HMDA	US	Y	Many non-racialized applicants experience consistently higher mortgage application denial rates when compared to white, non-Hispanic opposite-sex couples.
Kou J.B. (2012)	Mortgage loan	Race / ethnicity	Mortgage and borrower characteristics by several banks in Miami, Florida	US	Y	Borrowers in predominantly black neighborhoods pay a significantly higher contract rate than is consistent with evidence of their behavior.
Wu Y. (2012)	Farm loan	Race / ethnicity; Gender	Loan application database of the Georgia FSA State office	US	N	Approved loan amounts gap between racial and gender classes favors white and female borrowers; these borrower groups significantly dominate their peers in their financial strengths and greater capability to repay thus justifying favorable terms.
Agier I. (2013)	Microfinance	Gender	VivaCred a non-profit Brazil MFI	Brazil	Y	No gender bias in loan denial, but reveals disparate treatment with regard to credit conditions, particularly a "glass ceiling" effect where female applicants are less likely to be approved for a borrower's project goals.

TABLE A.3 Identification of discrimination studies, includes whether overall unwarranted discrimination found, research approach used and regional distribution.

Authors (Year)	Credit Application	Protected characteristic of interest	Data Used	Country	Unwarranted discrimination (Y/N)	Key Findings
Barasinska M. (2014)	P2P lending	Gender	German platform Snava	Ger many	N	No effect of gender on the individual borrower's chance to receive funds on this platform; female discrimination eased by the wisdom of the lending crowd.
Diaz-Serrano L. (2014)	Mortgage loan	Nationality / immigration status	Undisclosed real estate company	Spain	Y	Immigrant borrowers are charged substantially higher interest rates in their mortgages than their native counterparts.
Mason DR. (2014)	Microfinance	Gender	Microcredit clients from branch offices in six regions	Nicaragua	Y	Borrower assets, gender and length of time with the lending institution influence the size of loans received.
Cheng P. (2015)	Mortgage loan	Race / ethnicity; Gender; Age	US Survey of Consumer Finances	US	Y	Black borrowers pay 29 bps more than comparable white borrowers with rate disparity occurring to young borrowers with no education. Black women receive much more discrimination than black men.
Courchane M. (2015)	Mortgage loan	Race / ethnicity	Proprietary loan level records from representative mortgage lenders	US	Y	While the gender wage gap is well documented, there is no clear evidence for discrimination against other minority groups, after accounting for loan-level characteristics. Asians have denial rates as high as other minorities.
Doku S.Y. (2016)	General loan	Race / ethnicity	Living Costs and Food Survey	UK	Y	Non-white households are less likely to have financing compared to white households and even if they obtain financing, the intensity of borrowing is lower.
Tzoumis K. (2016)	Mortgage loan	Race / ethnicity	Subprime lender	US	Y	Pricing differentials for black applicants fluctuates substantially across time, even across relatively short time periods indicating possible dynamics in aggregate discrimination patterns.
Chen D. (2017)	P2P lending	Gender	Unnamed P2P lending platform in China	China	Y	Female borrowers are more likely to be funded than male borrowers, however, pay higher interest rates in exchange for higher funding success implying profit-based statistical discrimination and costly taste-based discrimination co-exist.
Corsi M. (2017)	Microfinance	Gender	Wekembe	Uganda	N	No evidence of gender discrimination against women clients, even though our results show that the loan size is influenced by personal characteristics and that women, in contrast to men, are rewarded according to their credit history.
Kara A. (2017)	Mortgage loan	Race / ethnicity	Living Costs and Food Survey	UK	N	Average probability of obtaining a mortgage is similar for White and Non-White households; however, Black households with low incomes are less likely to have mortgages whereas Asian households do not have a lower probability.
Seethi D. (2017)	Mortgage loan	Nationality / immigration status	City Hospital of Varese	Italy	Y	Discrimination is likely to appear when the applicant's nationality is considered.
Clarke K.A. (2018)	Mortgage loan	Race / ethnicity	Large national mortgage company	US	Y	Minority borrowers pay more on average for mortgages than non-minorities and that this effect persists even in the presence of unmeasured confounders.
Koplin N. (2018)	Mortgage loan	Race / ethnicity	GSS Sensitive Data Files: HMDA	US	Y	Blacks are not only more likely to be denied conventional home mortgages but denial rates among blacks are geographically correlated with racial prejudice, particularly among first-time home purchase loans and loans from depository lenders.
Smith B.C. (2018)	Mortgage loan	Race / ethnicity	LPS Analytics loan data	US	Y	Borrowers with comparable credit quality experience significantly higher costs for mortgages in neighborhoods with a high density of minority households.
Dols M.D. (2019)	Mortgage loan	Race / ethnicity	HMDA	US	N	Very little discrimination in loan origination, although discrimination increases somewhat after 2007. Its probability remains well below 1%.
Dhakar CK. (2019)	Farm loan	Race / ethnicity; Gender	USDA- FSA	US	Y	Non-white farm borrowers tend to receive larger loans among lower loan latent class, but receive relatively lower loans in larger loans borrower category. Male borrowers are accommodated with larger loans and longer maturities in all classes.
Hauptert (2019)	Mortgage loan	Race / ethnicity	HMDA	US	Y	The likelihood of a borrower of any race receiving subprime terms increased as the proportion of white residents in neighborhoods decreased, however, Asian borrowers experience favorable outcomes in home-generally nonwhite neighborhoods.
Kumar S.M. (2019)	Farm loan	Caste	Second round of the India Human Development Survey	India	Y	Significant inter-caste differences in application rates, and evidence of discrimination against Scheduled Tribe borrowers at the approval stage, such that evidence for caste-based discrimination persists.
Sun H. (2019)	Mortgage loan	Marriage status / sexual orientation	HMDA; Fannie Mae Loan Performance data	US	Y	Lenders charge 0.02-0.2% higher interest to same-sex borrowers, which is equivalent to an annual total of \$8.6 to \$86.6 million in additional interest fees nationwide, meaning, like same-sex borrowers are less risky overall.
Chen S.Y. (2020)	P2P lending	Gender; Age; Education	Renren dai	China	Y	P2P market correctly uses education level of borrowers to evaluate creditworthiness; younger female borrowers, however, are less likely to be funded, even with lower probability of default.
Chen X. (2020)	P2P lending	Gender	Renren dai	China	Y	Lending to female borrowers is associated with better loan performance than their male peers; however, female borrowers have to compensate lenders by providing higher loan termination patterns, after controlling for the probability of a borrower defaulting or prepaying, female borrowers pay higher contract rates in the subprime mortgage market.
Fang L. (2020)	Mortgage loan	Gender	GMAC Residential Capital Company	US	Y	Borrowers of different genders have different loan termination patterns, after controlling for the probability of a borrower defaulting or prepaying, female borrowers pay higher contract rates in the subprime mortgage market.
Gonzalez Martiñez R. (2020)	General loan	Race / ethnicity; Gender	70 invited MFI credit officers	Bolivia	Y	Nonwhite women have double the chance of loan approval, but indigenous women have only 1.5 times the chance compared with non, suggesting positive caste-based discrimination favorable for nonwhite women in Bolivia.
Kashian R. (2020)	Mortgage loan	Race / ethnicity	HMDA	US	Y	Loan acceptance rates are adversely influenced by high minority locations, although the effect is larger for minority applicants; results support the hypothesis that community banks will yield higher rates of loan acceptance.
Kim D. (2020)	P2P lending	Gender; Age	Three platforms: Moneyauktion, Popfund, and 8percent	Korea	N	No gender discrimination found in three Korean P2P lending markets; borrower's gender does not have a statistically significant impact on the lenders' preferences as well as his/her actual repayment performance.
Bertrand J. (2021)	P2P lending	Race / ethnicity	Unnamed P2P lender	US	Y	Persistence of discrimination in P2P lending against African American borrowers with higher loan rejection, higher loan spread, and shorter loan maturity in areas with higher proportions of African American residents.
Li Y. (2021)	Car loan	Gender	Portfolios of used car loans by financial institution	India	Y	Taste-based discrimination against women with low CIBIL scores exists where a female applicant must obtain a spectacular CIBIL score to eliminate the negative consequences of gender inequality.
Linskey-Talifero D. (2021)	Mortgage loan	Race / ethnicity; Gender; Age	HMDA	US	Y	Black and Hispanic denial odds for reverse mortgages were 107% and 48% higher than for non-Hispanic white applicants, denial odds for women were 1.2% lower than for men.
Butler A.W. (2022)	Car loan	Race / ethnicity	Credit bureau records; HMDA	US	Y	Black and Hispanic applicants' approval rates are 1.5 percentage points lower with minority borrowers paying 70 basis-points higher interest rates, but less severe disparities, indicating racial bias rather than statistical discrimination.
Loya (2022)	Mortgage loan	Race / ethnicity; Marriage status / sexual orientation	HMDA	US	Y	Latin American borrowers with higher loan rejection, higher loan spread, and shorter loan maturity in areas with higher proportions of African American residents.
Tsai J.C. (2022)	Mortgage loan	Gender	Federal Home Loan Bank	US	N	Women's status tilted from disadvantageous to advantageous; female borrowers in the highest income group received their lowest interest rates in 2015-2017. 0.026 percentage points lower than their male peers.

TABLE A.4 Identification of discrimination studies, includes whether overall unwarranted discrimination found, research approach used and regional distribution.

references

- [1] Marron D. Consumer Credit in the United States: A Sociological Perspective from the 19th Century to the Present. Palgrave Macmillan; 2009.
- [2] Castelnovo A, Malandri L, Mercurio F, Mezzanzanica M, Cosentini A. Towards Fairness Through Time. Communications in Computer and Information Science 2021;1524 CCIS:647–663. https://link.springer.com/chapter/10.1007/978-3-030-93736-2_46.
- [3] Kirkinis K, Pieterse AL, Martin C, Agiliga A, Brownell A. Racism, racial discrimination, and trauma: a systematic review of the social science literature. <https://doi.org/10.1080/1355785820181514453> 2018;26:392–412. <https://www.tandfonline.com/doi/abs/10.1080/13557858.2018.1514453>.
- [4] Mariani MM, Machado I, Magrelli V, Dwivedi YK. Artificial intelligence in innovation research: A systematic review, conceptual framework, and future research directions. Technovation 2022 9;p. 102623.
- [5] Shi S, Tse R, Luo W, D'Addona S, Pau G. Machine learning-driven credit risk: a systemic review. Neural Computing and Applications 2022 9;34:14327–14339. <https://link.springer.com/article/10.1007/s00521-022-07472-2>.
- [6] Tranfield D, Denyer D, Smart P. Towards a Methodology for Developing Evidence-Informed Management Knowledge by Means of Systematic Review. British Journal of Management 2003;14:207–222.
- [7] Vrontis D, Christofi M. RD internationalization and innovation: A systematic review, integrative framework and future research directions. Journal of Business Research 2021 5;128:812–823.
- [8] Bolt EET, Winterton J, Cafferkey K. A century of labour turnover research: A systematic literature review. International Journal of Management Reviews 2022 10;24:555–576.
- [9] Larsson S. The socio-legal relevance of artificial intelligence. Droit et société 2019;103:573–593. https://www.cairn-int.info/article.php?ID_ARTICLE=E_DRS1_103_0573.
- [10] Office of the Executive President, Big data: A report on algorithmic systems, opportunity, and civil rights; 2016. https://obamawhitehouse.archives.gov/sites/default/files/microsites/ostp/2016_0504_data_discrimination.pdf.
- [11] Select Committee on Artificial Intelligence. AI in the UK: Ready, willing and able? House of Lords Report 2018;<http://www.parliament.uk/mps-lords-and-offices/standards-and-interests/register-of-lords->.
- [12] High-Level Expert Group on Artificial Intelligence, Ethics guidelines for trustworthy AI; 2019. <https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>.
- [13] Heikkiläarchive M. The EU wants to put companies on the hook for harmful AI. MIT Technology Review 2022;<https://www.technologyreview.com/2022/10/01/1060539/eu-tech-policy-harmful-ai-liability/>.
- [14] OSTP, Blueprint for an AI Bill of Rights; 2022. <https://www.whitehouse.gov/ostp/ai-bill-of-rights/>.
- [15] Wyden R. Wyden, Booker and Clarke Introduce Algorithmic Accountability Act of 2022 To Require New Transparency And Accountability For Automated Decision Systems. US Senator Ron Wyden of Oregon 2022;<https://www.wyden.senate.gov/news/press-releases/wyden-booker-and-clarke-introduce-algorithmic-accountability-act-of-2022-to-require-new-transparency-and-accountability-for-automated-decision-systems?peek=BH793HGzEX7gimi20t7HiHEg8n9b3vET476N7MsTy%2BcOuyHe>.
- [16] Munnell AH, Tootell GMB, Browne LE, McEneaney J. Mortgage Lending in Boston: Interpreting HMDA Data on JSTOR. The American Economic Review 1996;86:25–53. https://www.jstor.org/stable/2118254#metadata_info_tab_contents.
- [17] Pope DG, Sydnor JR. What's in a Picture? Evidence of Discrimination from Prosper. com. Journal of Human resources 2011;46:53–92.

- [18] Hajian S, Domingo-Ferrer J. A methodology for direct and indirect discrimination prevention in data mining. *IEEE Transactions on Knowledge and Data Engineering* 2013;25:1445–1459.
- [19] Reamer A. 2021 Home Mortgage Disclosure Act (HMDA) datasets released by FFIEC. American Economic Association 2022; <https://www.aeaweb.org/forum/2701/2021-home-mortgage-disclosure-hmda-datasets-released-ffiec>.
- [20] Mehrabi N, Morstatter F, Saxena N, Lerman K, Galstyan A. A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys (CSUR)* 2021 7;54. <https://dl.acm.org/doi/10.1145/3457607>.
- [21] Zheng H, Chen Z, Du T, Zhang X, Cheng Y, Ti S, et al. NeuronFair: Interpretable White-Box Fairness Testing through Biased Neuron Identification. 2022 IEEE/ACM 44th International Conference on Software Engineering (ICSE) 2022;p. 1519–1531.
- [22] Kumar SM. Does Access to Formal Agricultural Credit Depend on Caste? *World Development* 2013 3;43:315–328.
- [23] Kumar SM, Venkatachalam R. Caste and Credit: A Woeful Tale? *The Journal of Development Studies* 2019 8;55:1816–1833. <https://www.tandfonline.com/doi/abs/10.1080/00220388.2018.1425397>.
- [24] Donthu N, Kumar S, Mukherjee D, Pandey N, Lim WM. How to conduct a bibliometric analysis: An overview and guidelines. *Journal of Business Research* 2021 9;133:285–296.
- [25] Jia H, Zhou S, Allaway AW. Understanding the evolution of consumer psychology research: A bibliometric and network analysis. *Journal of Consumer Behaviour* 2018 9;17:491–502. <https://onlinelibrary.wiley.com/doi/full/10.1002/cb.1734> <https://onlinelibrary.wiley.com/doi/abs/10.1002/cb.1734> <https://onlinelibrary.wiley.com/doi/10.1002/cb.1734>.
- [26] van Eck NJ, Waltman L. Software survey: VOSviewer, a computer program for bibliometric mapping. *Scientometrics* 2010 12;84:523–538. <https://link.springer.com/article/10.1007/s11192-009-0146-3>.
- [27] Gotham KF. Race, Mortgage Lending and Loan Rejections in a U.S. City. *Sociological Focus* 1998;31:391–405. <https://www.tandfonline.com/doi/abs/10.1080/00380237.1998.10571116>.
- [28] Ards SD, Ha IS, Mazas JL, Myers SL. Bad Credit and Intergroup Differences in Loan Denial Rates. *The Review of Black Political Economy* 2015 1;42:19–34. <https://journals.sagepub.com/doi/10.1007/s12114-014-9179-9>.
- [29] Avery RB, Brevoort KP, Canner G. Does Credit Scoring Produce a Disparate Impact? *Real Estate Economics* 2012 12;40:65–114. <https://ideas.repec.org/a/bla/reesec/v40y2012ips65-s114.html>.
- [30] Bernerth JB. Demographic Variables and Credit Scores: An empirical study of a controversial selection tool. *International Journal of Selection and Assessment* 2012 6;20:242–246. <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1468-2389.2012.00585.x> <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-2389.2012.00585.x>.
- [31] Goodman LS, Bai B, Li W. Real Denial Rates: A New Tool to Look at Who Is Receiving Mortgage Credit. *Housing Policy Debate* 2019;29:795–819. <https://www.tandfonline.com/doi/abs/10.1080/10511482.2018.1524441>.
- [32] Black H, Schweitzer RL, Mandell L. Discrimination in Mortgage Lending. *American Economic Review* 1978;68:186–91.
- [33] Horne DK. Evaluating the Role of Race in Mortgage Lending. *FDIC Banking Review* 1994;7.
- [34] Glennon D, Stengel M. An Evaluation of the Federal Reserve Bank of Boston's Study of Racial Discrimination in Mortgage Lending. *Economic and Policy Analysis Working Paper 94-2*, Office of the Comptroller of the Currency 1994;.
- [35] Horne DK. Mortgage Lending, Race, and Model Specification. *Journal of Financial Services Research* 1997;11:43–68.
- [36] Liebowitz S. A Study That Deserves No Credit. *The Wall Street Journal* 1993 1;.

- [37] Berkovec JA, Canner GB, Gabriel SA, Hannan TH. Race, redlining, and residential mortgage loan performance. *The Journal of Real Estate Finance and Economics* 1994 11;9:263–294. <https://link.springer.com/article/10.1007/BF01099279>.
- [38] Ingram FJ, Warner AE. A General Test for Compliance with Equal Credit Opportunity Standards. *Journal of Consumer Affairs* 1981 12;15:232–248. <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1745-6606.1981.tb00711.xhttps://onlinelibrary.wiley.com/doi/abs/10.1111/j.1745-6606.1981.tb00711.xhttps://onlinelibrary.wiley.com/doi/10.1111/j.1745-6606.1981.tb00711.x>.
- [39] McMurray JP, Thomson TA. The Influence of Race in Residential Mortgage Closings. *Journal of Real Estate Research* 2003;25:543–556. <https://ideas.repec.org/a/jre/issued/v25n42003p543-556.html>.
- [40] Guy RF, Pol LG, Ryker RE. Discrimination in mortgage lending: The Home Mortgage Disclosure Act. *Population Research and Policy Review* 1982 10;1:283–296. <https://link.springer.com/article/10.1007/BF00140097>.
- [41] Leahy PJ. Are Racial Factors Important for the Allocation of Mortgage Money? *American Journal of Economics and Sociology* 1985 4;44:185–196. <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1536-7150.1985.tb02333.xhttps://onlinelibrary.wiley.com/doi/abs/10.1111/j.1536-7150.1985.tb02333.xhttps://onlinelibrary.wiley.com/doi/10.1111/j.1536-7150.1985.tb02333.x>.
- [42] Myers SL, Chan T. Racial discrimination in housing markets: Accounting for credit risk. *Social Science Quarterly* 1995;76:543–561. <https://psycnet.apa.org/record/1996-17023-001>.
- [43] Black HA, Collins MC, Cyree KB. Do Black-Owned Banks Discriminate against Black Borrowers? *Journal of Financial Services Research* 1997;11:189–204. <https://link.springer.com/article/10.1023/A:1007943610378>.
- [44] Blackburn M, Vermilyea T. A Comparison of Unexplained Racial Disparities in Bank-Level and Market-Level Models of Mortgage Lending. *Journal of Financial Services Research* 2006 3;29:125–147. <https://link.springer.com/article/10.1007/s10693-006-5922-0>.
- [45] Dietrich J. Classifying applicants for fair lending analyses: What do the data have to say? *Journal of Economic and Social Measurement* 2010 1;35:231–253.
- [46] Megbolugbe IF. Understanding Mortgage Lending Patterns. *Journal of Housing Research* 1993;4:185–189. <https://www.jstor.org/stable/24832774>.
- [47] Squires GD, Vélez W. Mortgage Lending and Race: Is Discrimination Still a Factor? *Environment and Planning A: Economy and Space* 1996 7;28:1199–1208. <https://journals.sagepub.com/doi/10.1068/a281199>.
- [48] Holloway SR. Exploring the Neighborhood Contingency of Race Discrimination in Mortgage Lending in Columbus, Ohio. *Annals of the Association of American Geographers* 1998;88:252–276. <https://www.tandfonline.com/doi/abs/10.1111/1467-8306.00093>.
- [49] Williams RA, Nesiba RF. Racial, economic, and institutional differences in home mortgage loans: St. Joseph County, Indiana. *Journal of Urban Affairs* 1997 3;19:73–103. <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1467-9906.1997.tb00398.xhttps://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9906.1997.tb00398.xhttps://onlinelibrary.wiley.com/doi/10.1111/j.1467-9906.1997.tb00398.x>.
- [50] Reibel M. Geographic variation in mortgage discrimination: Evidence from Los Angeles. *Urban Geography* 2000 1;21:45–60. <https://www.tandfonline.com/doi/abs/10.2747/0272-3638.21.1.45>.
- [51] Ross SL, Turner MA, Godfrey E, Smith RR. Mortgage lending in Chicago and Los Angeles: A paired testing study of the pre-application process. *Journal of Urban Economics* 2008 5;63:902–919.
- [52] Blank EC, Venkatachalam P, McNeil L, Green RD. Racial discrimination in mortgage lending in Washington, D.C.: A mixed methods approach. *The Review of Black Political Economy* 2005;33:9–30. <https://link.springer.com/article/10.1007/s12114-005-1012-z>.

- [53] Kopkin N. The conditional spatial correlations between racial prejudice and racial disparities in the market for home loans. *Urban Studies* 2018 3;55:3596–3614. <https://journals.sagepub.com/doi/abs/10.1177/0042098018755086>.
- [54] Delis MD, Papadopoulos P. Mortgage Lending Discrimination Across the U.S.: New Methodology and New Evidence. *Journal of Financial Services Research* 2019 12;56:341–368. <https://link.springer.com/article/10.1007/s10693-018-0290-0>.
- [55] Hauptert T. Racial Patterns in Mortgage Lending Outcomes During and After the Subprime Boom. *Housing Policy Debate* 2019 11;29:947–976. <https://www.tandfonline.com/doi/abs/10.1080/10511482.2019.1636286>.
- [56] Sun H, Gao L. Lending practices to same-sex borrowers. *Proceedings of the National Academy of Sciences of the United States of America* 2019 5;116:9293–9302. <https://pubmed.ncbi.nlm.nih.gov/30992376/>.
- [57] Lindsey-Taliefero D, Kelly L. Reverse Mortgage Lending Disparities and the Economically Vulnerable. *International Advances in Economic Research* 2021 8;27:159–169. <https://link.springer.com/article/10.1007/s11294-021-09831-6>.
- [58] Loya J. Ethno-racial stratification in the mortgage market: The role of co-applicants. *Social Science Research* 2022 8;106:102725.
- [59] Goering JM, Wienk RE. Mortgage lending, racial discrimination, and federal policy. Urban Institute Press; 1996.
- [60] Chen J, Kallus N, Mao X, Svacha G, Udell M. Fairness under unawareness: Assessing disparity when protected class is unobserved. *FAT* 2019 - Proceedings of the 2019 Conference on Fairness, Accountability, and Transparency* 2019 1;p. 339–348.
- [61] Hanson A, Hawley Z, Martin H, Liu B. Discrimination in mortgage lending: Evidence from a correspondence experiment. *Journal of Urban Economics* 2016 3;92:48–65.
- [62] Fienberg SE, Krislov SH, Straf ML. Understanding and evaluating statistical evidence in litigation. *Jurimetrics* 1995;36:1–32. <https://www.jstor.org/stable/29762401>.
- [63] Martin RE, Hill RC. Loan performance and race. *Economic Inquiry* 2000 1;38:136–150. <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1465-7295.2000.tb00009.x>
<https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1465-7295.2000.tb00009.x>
- [64] Escalante CL, Epperson JE, Raghunathan U. Gender Bias Claims in Farm Service Agency's Lending Decisions on JSTOR. *Journal of Agricultural and Resource Economics* 2009;34:332–349. https://www.jstor.org/stable/41548417#metadata_info_tab_contents.
- [65] Wu Y, Escalante CL, Gunter LF, Epperson JE. A decomposition approach to analysing racial and gender biases in Farm Service Agency's lending decisions. *Applied Economics* 2012 8;44:2841–2850. <https://www.tandfonline.com/doi/abs/10.1080/00036846.2011.566210>.
- [66] Dhakal CK, Escalante CL, Dodson C. Heterogeneity of farm loan packaging term decisions: a finite mixture approach. *Applied Economics Letters* 2019 10;26:1528–1532. <https://www.tandfonline.com/doi/abs/10.1080/13504851.2019.1584360>.
- [67] Barasinska N, Schäfer D. Is Crowdfunding Different? Evidence on the Relation between Gender and Funding Success from a German Peer-to-Peer Lending Platform. *German Economic Review* 2014 11;15:436–452. <https://onlinelibrary.wiley.com/doi/full/10.1111/geer.12052>
<https://onlinelibrary.wiley.com/doi/abs/10.1111/geer.12052>
- [68] Chen X, Huang B, Ye D. Gender gap in peer-to-peer lending: Evidence from China. *Journal of Banking Finance* 2020 3;112:105633. <https://linkinghub.elsevier.com/retrieve/pii/S0378426619302080>.

- [69] Kim D. Sexism and Ageism in a P2P Lending Market: Evidence from Korea. *The Journal of Asian Finance, Economics and Business* 2020;7:537–550. <http://Creativecommons.org/licenses/by-nc/4.0/>.
- [70] Zhao H, Ge Y, Liu Q, Wang G, Chen E, Zhang H. P2P lending survey: Platforms, recent advances and prospects. *ACM Transactions on Intelligent Systems and Technology (TIST)* 2017 7;8. <https://dl.acm.org/doi/10.1145/3078848>.
- [71] Li J, Hu J. Does university reputation matter? Evidence from peer-to-peer lending. *Finance Research Letters* 2019 12;31:66–77.
- [72] Caglayan M, Talavera O, Xiong L, Zhang J. What does not kill us makes us stronger: the story of repetitive consumer loan applications. *The European Journal of Finance* 2022;28:46–65. <https://www.tandfonline.com/doi/abs/10.1080/1351847X.2020.1793792>.
- [73] Chen D, Li X, Lai F. Gender discrimination in online peer-to-peer credit lending: evidence from a lending platform in China. *Electronic Commerce Research* 2017 12;17:553–583. <https://link.springer.com/article/10.1007/s10660-016-9247-2>.
- [74] Bertrand J, Weill L. Do algorithms discriminate against African Americans in lending? *Economic Modelling* 2021 11;104:105619.
- [75] Butler AW, Mayer EJ, Weston JP. Racial Disparities in the Auto Loan Market. *The Review of Financial Studies* 2022 5; <https://academic.oup.com/rfs/advance-article/doi/10.1093/rfs/hhac029/6588701>.
- [76] Li Y. Gender differences in car loan access: An empirical analysis. *ACM International Conference Proceeding Series* 2021 7;p. 493–498.
- [77] Lewis EM. *An Introduction to Credit Scoring*. Athena Press; 1994.
- [78] Boot AWA. Relationship Banking: What Do We Know? *Journal of Financial Intermediation* 2000 1;9:7–25.
- [79] Guryan J, Charles KK. Taste-Based or Statistical Discrimination: The Economics of Discrimination Returns to its Roots. *The Economic Journal* 2013 11;123:F417–F432. <https://academic.oup.com/ej/article/123/572/F417/5080752>.
- [80] Albareto G, Galardo M, Mistrulli PE, Sorvillo B. Culture, Lending Relationships, and the Cost of Credit. *The Review of Corporate Finance Studies* 2022 7;11:736–774. <https://academic.oup.com/rcfs/article/11/3/736/6513393>.
- [81] Agier I, Szafarz A. Subjectivity in credit allocation to micro-entrepreneurs: Evidence from Brazil. *Small Business Economics* 2013 6;41:263–275. <https://link.springer.com/article/10.1007/s11187-012-9429-9>.
- [82] de Janvry A, McIntosh C, Sadoulet E. The supply- and demand-side impacts of credit market information. *Journal of Development Economics* 2010 11;93:173–188.
- [83] Hossein CS. The Exclusion of Afro-Guyanese Hucksters in Micro-Banking. *European Review of Latin American and Caribbean Studies* 2014;p. 75–98. https://www.jstor.org/stable/23722435#metadata_info_tab_contents.
- [84] Harkness SK. Discrimination in Lending Markets: Status and the Intersections of Gender and Race. *Social Psychology Quarterly* 2016 2;79:81–93. <https://journals.sagepub.com/doi/abs/10.1177/0190272515623459>.
- [85] Thomas LC, Edelman DB, Crook JN. *Credit scoring and its applications*. SIAM; 2002.
- [86] Siddiqi N. *Credit risk scorecards: Developing and implementing intelligent credit scoring* 2013;.
- [87] Anderson R. *The credit scoring toolkit: Theory and practice for retail credit risk management and decision automation*. Oxford University Press; 2007.
- [88] Han S. Creditor Learning and Discrimination in Lending. *Journal of Financial Services Research* 2011 10;40:1–27. <https://link.springer.com/article/10.1007/s10693-011-0101-3>.

- [89] Afolabi AO, Akinlolu IR. Evaluation of women's access to building credits from banks in Nigeria. *Banks and Bank Systems* 2021;16:45–60.
- [90] Ghosh S, Vinod D. What constrains financial inclusion for women? Evidence from Indian micro data. *World Development* 2017;92:60–81.
- [91] Huang W, Qian Y, Xu N. The signaling effects of education in the online lending market: Evidence from China. *Economic Modelling* 2020 11;92:268–276. <https://linkinghub.elsevier.com/retrieve/pii/S0264999319318589>.
- [92] Rao S, Malapit HJL. Gender, Household Structure and Financial Participation in the United States. *Journal of Family and Economic Issues* 2015 12;36:606–620. <https://link.springer.com/article/10.1007/s10834-014-9426-z>.
- [93] Robinson JK. Race, Gender, and Familial Status: Discrimination in One US Mortgage Lending Market. *Feminist Economics* 2002 7;8:63–85. <https://www.tandfonline.com/doi/abs/10.1080/13545700210167323>.
- [94] Nyarko SA. Gender discrimination and lending to women: The moderating effect of an international founder. *International Business Review* 2022 8;31:101973.
- [95] Kulik CT, Holbrook RL. Demographics in Service Encounters: Effects of Racial and Gender Congruence on Perceived Fairness. *Social Justice Research* 2000;13:375–402. <https://link.springer.com/article/10.1023/A:1007675010798>.
- [96] Bullock HE, Toolis EE, Sencion B, Cadenas MT. The high price of economic marginalization: Low-income Latinas' experiences with mainstream banking and alternative financial services. *Peace and Conflict: Journal of Peace Psychology* 2020 5;26:136–148. /record/2020-30003-005.
- [97] Goenner CF. Majority-Minority Boards of Directors and Decision Making: The Effects of Homophily on Lending Decisions. *Business Society* 2022 12; <https://journals.sagepub.com/doi/full/10.1177/00076503211062182>.
- [98] Squires GD, Kim S. Does Anybody Who Works Here Look Like Me: Mortgage Lending, Race, and Lender Employment on JSTOR. *Social Science Quarterly* 1995;76:823–838. https://www.jstor.org/stable/44072686#metadata_info_tab_contents.
- [99] Fu R, Huang Y, Singh PV. Crowds, Lending, Machine, and Bias. *Information Systems Research* 2021 2;32:72–92. <https://pubsonline.informs.org/doi/abs/10.1287/isre.2020.0990>.
- [100] Bocian DG, Ernst KS, Li W. Race, ethnicity and subprime home loan pricing. *Journal of Economics and Business* 2008 1;60:110–124.
- [101] Fang L, Munneke HJ. Gender Equality in Mortgage Lending. *Real Estate Economics* 2020 12;48:957–1003. <https://onlinelibrary.wiley.com/doi/full/10.1111/1540-6229.12198> <https://onlinelibrary.wiley.com/doi/abs/10.1111/1540-6229.12198> <https://onlinelibrary.wiley.com/doi/10.1111/1540-6229.12198>.
- [102] Boehm TP, Thistle PD, Schlottmann A. Rates and race: An analysis of racial disparities in mortgage rates. *Housing Policy Debate* 2006;17:109–149. <https://www.tandfonline.com/doi/abs/10.1080/10511482.2006.9521563>.
- [103] Courchane MJ. The Pricing of Home Mortgage Loans to Minority Borrowers: How Much of the APR Differential Can We Explain? *Journal of Real Estate Research* 2007 10;29:399–439. <https://www.tandfonline.com/doi/abs/10.1080/10835547.2007.12091208>.
- [104] Crawford GW, Rosenblatt E. Differences in the Cost of Mortgage Credit Implications for Discrimination. *The Journal of Real Estate Finance and Economics* 1999;19:147–159. <https://link.springer.com/article/10.1023/A:1007831506399>.
- [105] Cheng P, Lin Z, Liu Y. Do Women Pay More for Mortgages? *Journal of Real Estate Finance and Economics* 2011 11;43:423–440. https://www.researchgate.net/publication/225702859_Do_Women_Pay_More_for_Mortgages.

- [106] Diaz-Serrano L, Raya JM. Mortgages, immigrants and discrimination: An analysis of the interest rates in Spain. *Regional Science and Urban Economics* 2014 3;45:22–32.
- [107] Tsai IC. Have men and women become equal in the housing market? Effects of gender on mortgage rate. *Journal of Housing and the Built Environment* 2022 4; <https://link.springer.com/article/10.1007/s10901-022-09944-2>.
- [108] Clarke KA, Rothenberg LS. Mortgage Pricing and Race: Evidence from the Northeast. *American Law and Economics Review* 2018 4;20:138–167. <https://academic.oup.com/aler/article/20/1/138/4601874>.
- [109] Collins MC, Harvey KD, Nigro PJ. Mortgage Broker Loan Pricing Leading Up to the Financial Crisis: Were Yield Spread Premiums the Only Problem? *Housing Policy Debate* 2021;31:875–890. <https://www.tandfonline.com/doi/abs/10.1080/10511482.2020.1862892>.
- [110] Kau J, Keenan D, Munneke H. Racial Discrimination and Mortgage Lending. *The Journal of Real Estate Finance and Economics* 2012 8;45:289–304. <https://EconPapers.repec.org/RePEc:kap:jrefec:v:45:y:2012:i:2:p:289-304>.
- [111] Kau JB, Fang L, Munneke HJ. An Unintended Consequence of Mortgage Financing Regulation – a Racial Disparity. *Journal of Real Estate Finance and Economics* 2019 11;59:549–588. <https://link.springer.com/article/10.1007/s11146-018-9683-y>.
- [112] Hodge ME, Dawkins MC, Reeves JH. A Case Study of Mortgage Refinancing Discrimination: African American Inter-generational Wealth. *Sociological Inquiry* 2007 2;77:23–43. <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1475-682x.2007.00176.x> <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1475-682x.2007.00176.x> <https://onlinelibrary.wiley.com/doi/10.1111/j.1475-682x.2007.00176.x>.
- [113] Avery RB, Brevoort KP, Canner GB. Higher-priced home lending and the 2005 HMDA data. *Federal Reserve Bulletin* 2006;92:123–166. <https://ideas.repec.org/a/fip/fedgrb/y2006iseppa123-a166nv.92.html> <https://ideas.repec.org/a/fip/fedgrb/y2006iseppa123-a166nv.92.html>.
- [114] Black HA, Boehm TP, DeGennaro RP. Is there discrimination in mortgage pricing? The case of overages. *Journal of Banking Finance* 2003 6;27:1139–1165.
- [115] Berkovec JA, Canner GB, Gabriel SA, Hannan TH. Discrimination, Competition, and Loan Performance in FHA Mortgage Lending. *The Review of Economics and Statistics* 1998 5;80:241–250. <https://direct.mit.edu/rest/article/80/2/241/57067/Discrimination-Competition-and-Loan-Performance-in>.
- [116] Anyamele OD. Racial/Ethnic Differences in Household Loan Delinquency Rate. *The Review of Black Political Economy* 2015 1;42:415–442. <https://journals.sagepub.com/doi/10.1007/s12114-015-9213-6>.
- [117] Harrison DM. The Importance of Lender Heterogeneity in Mortgage Lending. *Journal of Urban Economics* 2001 3;49:285–309.
- [118] Bayer P, Ferreira F, Ross SL. What Drives Racial and Ethnic Differences in High-Cost Mortgages? The Role of High-Risk Lenders. *The Review of Financial Studies* 2018 1;31:175–205. <https://academic.oup.com/rfs/article/31/1/175/3782656>.
- [119] Dobbie W, Liberman A, Paravisini D, Pathania V. Measuring Bias in Consumer Lending. *The Review of Economic Studies* 2021 11;88:2799–2832. <https://academic.oup.com/restud/article/88/6/2799/6342424>.
- [120] Coffey B, Roberts C. Racial Differences in Mortgage Lending in the Southeastern United States. *Southeastern Geographer* 1999;39:46–60. <https://muse.jhu.edu/article/425505/summary>.
- [121] Fisman R, Sarkar A, Skrastins J, Vig V. Experience of Communal Conflicts and Intergroup Lending. *Journal of Political Economy* 2020 9;128:3346–3375. <https://www.journals.uchicago.edu/doi/10.1086/708856>.
- [122] Wheeler CH, Olson LM. Racial differences in mortgage denials over the housing cycle: Evidence from U.S. metropolitan areas. *Journal of Housing Economics* 2015 12;30:33–49.

- [123] Deku SY, Kara A, Smith K, Xia M. Ethnic minorities' access to mortgages in the UK: The undesirable impact of the Great Financial Crisis. *Finance Research Letters* 2022 3;45:102183.
- [124] Harvey KD, Collins MC, Nigro P, Robinson B. Disparities in Mortgage Lending, Bank Performance, Economic Influence, and Regulatory Oversight. *The Journal of Real Estate Finance and Economics* 2001;23:379–410. <https://link.springer.com/article/10.1023/A:1017908305060>.
- [125] Dillbary JS, Edwards G. An Empirical Analysis of Sexual Orientation Discrimination. *The University of Chicago Law Review* 2019;86. <http://perma.cc/CC58-DRYG>.
- [126] Antoniadis A, Marafi F, Antoniadis A, Marafi F. Can Jane Get a Mortgage Loan? Depends on When and Where. *Quantitative Finance and Economics* 2017;1:67–93. <http://www.aimspress.com/article/doi/10.3934/QFE.2017.1.67><http://www.aimspress.com/rcticle/doi/10.3934/QFE.2017.1.67>.
- [127] Tantri P. Fintech for the Poor: Financial Intermediation Without Discrimination. *Review of Finance* 2021 4;25:561–593. <https://academic.oup.com/rof/article/25/2/561/6028998>.
- [128] Kelley S, Ovchinnikov A, Hardoon DR, Heinrich A. Antidiscrimination Laws, Artificial Intelligence, and Gender Bias: A Case Study in Nonmortgage Fintech Lending. *Manufacturing Service Operations Management* 2022 5;p. 1–21. <https://pubsonline.informs.org/doi/abs/10.1287/msom.2022.1108>.
- [129] Bono T, Croxson K, Giles A. Algorithmic fairness in credit scoring. *Oxford Review of Economic Policy* 2021 9;37:585–617. <https://academic.oup.com/oxrep/article/37/3/585/6374682>.
- [130] Fuster A, Goldsmith-Pinkham P, Ramadorai T, Walther A. Predictably Unequal? The Effects of Machine Learning on Credit Markets. *Journal of Finance* 2022 2;77:5–47.
- [131] Hauptert T. The Racial Landscape of Fintech Mortgage Lending. *Housing Policy Debate* 2022;32:337–368. <https://www.tandfonline.com/doi/abs/10.1080/10511482.2020.1825010>.
- [132] Bartlett R, Morse A, Stanton R, Wallace N. Consumer-lending discrimination in the FinTech Era. *Journal of Financial Economics* 2022 1;143:30–56.
- [133] Futures F. Leveling the banking playing field for the masses; 2021. <https://www.fintechfutures.com/2021/06/leveling-the-banking-playing-field-for-the-masses/>.
- [134] Wooldridge JM. *Econometric analysis of cross section and panel data*. The MIT Press; 2010.
- [135] Lainez N. The Prospects and Dangers of Algorithmic Credit Scoring in Vietnam: Regulating a Legal Blindspot. ISEAS Yusof Ishak Institute 2021 1;<https://think-asia.org/handle/11540/13169>.
- [136] Barocas S, Selbst AD. Big data's disparate impact. *California Law Review* 2016;104:671–732.
- [137] Kleinberg J, Ludwig J, Mullainathan S, Rambachan A. Algorithmic Fairness. *AEA Papers and Proceedings* 2018 5;108:22–27.
- [138] Lipton ZC, Chouldechova A, McAuley J. Does mitigating ML's impact disparity require treatment disparity? *Advances in Neural Information Processing Systems* 2018;31.
- [139] Žliobaitė I, Custers B. Using sensitive personal data may be necessary for avoiding discrimination in data-driven decision models. *Artificial Intelligence and Law* 2016 6;24:183–201. <https://link.springer.com/article/10.1007/s10506-016-9182-5>.
- [140] Andreeva G, Matuszyk A. The law of equal opportunities or unintended consequences?: The effect of unisex risk assessment in consumer credit. *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 2019 10;182:1287–1311. <https://onlinelibrary.wiley.com/doi/full/10.1111/rssa.12494><https://onlinelibrary.wiley.com/doi/abs/10.1111/rssa.12494><https://rss.onlinelibrary.wiley.com/doi/10.1111/rssa.12494>.

- [141] Alufaisan Y, Kantarcioglu M, Zhou Y. Detecting Discrimination in a Black-Box Classifier. *Proceedings - 2016 IEEE 2nd International Conference on Collaboration and Internet Computing* 2017 1;p. 329–338.
- [142] Ruggieri S, Pedreschi D, Turini F. Integrating induction and deduction for finding evidence of discrimination. *Artificial Intelligence and Law* 2010 3;18:1–43. <https://link.springer.com/article/10.1007/s10506-010-9089-5>.
- [143] Pedreshi D, Ruggieri S, Turini F. Discrimination-aware data mining. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 2008;p. 560–568. <https://dl.acm.org/doi/10.1145/1401890.1401959>.
- [144] Berendt B, Preibusch S. Better decision support through exploratory discrimination-aware data mining: Foundations and empirical evidence. *Artificial Intelligence and Law* 2014 1;22:175–209. <https://link.springer.com/article/10.1007/s10506-013-9152-0>.
- [145] Zhang Y. Assessing Fair Lending Risks Using Race/Ethnicity Proxies. *Management Science* 2018 1;64:178–197. <http://pubsonline.informs.org/doi/10.1287/mnsc.2016.2579>.
- [146] Kallus N, Mao X, Zhou A. Assessing Algorithmic Fairness with Unobserved Protected Class Using Data Combination. *Management Science* 2021 6;<https://arxiv.org/abs/1906.00285v2>.
- [147] Fernando MP, Cèsar F, David N, José HO. Missing the missing values: The ugly duckling of fairness in machine learning. *International Journal of Intelligent Systems* 2021 7;36:3217–3258. <https://onlinelibrary.wiley.com/doi/full/10.1002/int.22415><https://onlinelibrary.wiley.com/doi/abs/10.1002/int.22415><https://onlinelibrary.wiley.com/doi/10.1002/int.22415>.
- [148] Corbett-Davies S, Pierson E, Feller A, Goel S, Huq A. Algorithmic decision making and the cost of fairness. vol. Part F1296; 2017. p. 797–806.
- [149] Goel N, Amayuelas A, Deshpande A, Sharma A, Zürich E, Lausanne E, et al. The Importance of Modeling Data Missingness in Algorithmic Fairness: A Causal Perspective. *35th AAAI Conference on Artificial Intelligence* 2021 12;p. 7564–7573. <https://arxiv.org/abs/2012.11448v1>.
- [150] Verma S, Rubin J. Fairness definitions explained. *IEEE Computer Society*; 2018. p. 1–7.
- [151] Hardt M, Price E, Price E, Srebro N. Equality of Opportunity in Supervised Learning. *Advances in Neural Information Processing Systems* 2016;29.
- [152] Berk R, Heidari H, Jabbari S, Kearns M, Roth A. Fairness in criminal justice risk assessments: The state of the art. *Sociological Methods and Research* 2017 3;50:3–44.
- [153] Dwork C, Hardt M, Pitassi T, Reingold O, Zemel R. Fairness Through Awareness. *ITCS 2012 - Innovations in Theoretical Computer Science Conference* 2011 4;p. 214–226. <https://arxiv.org/abs/1104.3913v2>.
- [154] Kusner M, Loftus J, Russell C, Silva R. Counterfactual Fairness. *Proceedings of the 31st International Conference on Neural Information Processing Systems* 2017;p. 4069–4079. <https://dl.acm.org/doi/10.5555/3294996.3295162>.
- [155] Chouldechova A, Roth A. The Frontiers of Fairness in Machine Learning 2018 10;<https://arxiv.org/abs/1810.08810v1>.
- [156] Chouldechova A. Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments. *Big Data* 2017 6;5:153–163. <https://www.liebertpub.com/doi/10.1089/big.2016.0047>.
- [157] Zafar MB, Valera I, Rodriguez MG, Gummadi KP. Fairness beyond disparate treatment disparate impact: Learning classification without disparate mistreatment. *26th International World Wide Web Conference, WWW 2017* 2017;p. 1171–1180. <https://dl.acm.org/doi/10.1145/3038912.3052660>.

- [158] Kleinberg J, Mullainathan S, Raghavan M. Inherent trade-offs in the fair determination of risk scores. *Leibniz International Proceedings in Informatics, LIPIcs* 2017 11;67.
- [159] Lundberg SM, Allen PG, Lee SI. A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems* 2017;30. <https://github.com/slundberg/shap>.
- [160] Hickey JM, Stefano PGD, Vasileiou V. Fairness by Explicability and Adversarial SHAP Learning. *Lecture Notes in Computer Science* 2021;12459 LNAI:174–190. https://link.springer.com/chapter/10.1007/978-3-030-67664-3_11.
- [161] Criado N, Ferrer X, Such JM. Attesting Digital Discrimination Using Norms. *International Journal of Interactive Multimedia and Artificial Intelligence* 2021;6:16–23. <http://www.bbc.co.uk/news/business-36129046>.
- [162] Balashankar A, Lees A. The Need for Transparent Demographic Group Trade-Offs in Credit Risk and Income Classification. *Lecture Notes in Computer Science* 2022;13192 LNCS:344–354. https://link.springer.com/chapter/10.1007/978-3-030-96957-8_30.
- [163] Lee NT, Lai S. Why New York City is cracking down on AI in hiring; 2021. <https://www.brookings.edu/blog/techtank/2021/12/20/why-new-york-city-is-cracking-down-on-ai-in-hiring/>.
- [164] Kozodoi N, Jacob J, Lessmann S. Fairness in credit scoring: Assessment, implementation and profit implications. *European Journal of Operational Research* 2022 3;297:1083–1094.
- [165] Gözl P, Kahng A, Procaccia AD. Paradoxes in fair machine learning. *Advances in Neural Information Processing Systems* 2019;32.
- [166] Friedler SA, Choudhary S, Scheidegger C, Hamilton EP, Venkatasubramanian S, Roth D. A comparative study of fairness-enhancing interventions in machine learning. *FAT* 2019 - Proceedings of the 2019 Conference on Fairness, Accountability, and Transparency* 2018 2;p. 329–338. <https://arxiv.org/abs/1802.04422v1>.
- [167] Iosifidis V, Ntoutsi E. Adafair: Cumulative fairness adaptive boosting. *International Conference on Information and Knowledge Management, Proceedings* 2019 11;p. 781–790.
- [168] Cai W, Gaebler J, Garg N, Goel S. Fair allocation through selective information acquisition. *AIES 2020 - Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* 2020 2;p. 22–28.
- [169] Agarwal C, Lakkaraju H, Zitnik M. Towards a Unified Framework for Fair and Stable Graph Representation Learning. *37th Conference on Uncertainty in Artificial Intelligence* 2021 2;p. 2114–2124. <https://arxiv.org/abs/2102.13186v3>.
- [170] Wagner B, d'Avila Garcez AS. Neural-symbolic integration for fairness in AI. *CEUR Workshop Proceedings* 2021;<http://ceur-ws.org/Vol-2846/paper5.pdf>.
- [171] Reich CL, Vijaykumar S. A Possibility in Algorithmic Fairness: Can Calibration and Equal Error Rates Be Reconciled? *Leibniz International Proceedings in Informatics, LIPIcs* 2021 2;192. <http://arxiv.org/abs/2002.07676><http://dx.doi.org/10.4230/LIPIcs.FORC.2021.4>.
- [172] Martinez N, Bertran M, Sapiro G. Minimax Pareto Fairness: A Multi Objective Perspective. *Proceedings of the 37th International Conference on Machine Learning* 2020;p. 6755–6764.
- [173] Gill N, Hall P, Montgomery K, Schmidt N. A Responsible Machine Learning Workflow with Focus on Interpretable Models, Post-hoc Explanation, and Discrimination Testing. *Information* 2020, Vol 11, Page 137 2020 2;11:137. <https://www.mdpi.com/2078-2489/11/3/137/html><https://www.mdpi.com/2078-2489/11/3/137>.
- [174] Biswas S, Rajan H. Do the Machine Learning Models on a Crowd Sourced Platform Exhibit Bias? An Empirical Study on Model Fairness. *Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering* 2020;<https://doi.org/10.1145/3368089.3409704>.

- [175] Stevens A, Deruyck P, Veldhoven ZV, Vanthienen J. Explainability and Fairness in Machine Learning: Improve Fair End-to-end Lending for Kiva. 2020 IEEE Symposium Series on Computational Intelligence 2020 12;p. 1241–1248.
- [176] Islam MT, Fariha A, Meliou A, Salimi B. Through the Data Management Lens: Experimental Analysis and Evaluation of Fair Classification. *Proceedings of the ACM SIGMOD International Conference on Management of Data 2022* 1;29:232–246. <https://arxiv.org/abs/2101.07361v4>.
- [177] Teodorescu MHM, Yao X. Machine Learning Fairness is Computationally Difficult and Algorithmically Unsatisfactorily Solved. 2021 IEEE High Performance Extreme Computing Conference 2021;.
- [178] Li Y, Ning Y, Liu R, Wu Y, Wang WH. Fairness of Classification Using Users' Social Relationships in Online Peer-To-Peer Lending. *The Web Conference 2020 - Companion of the World Wide Web Conference, WWW 2020* 2020 4;p. 733–742.
- [179] Holstein K, Vaughan JW, Daumé H, Dudík M, Wallach H. Improving fairness in machine learning systems: What do industry practitioners need? *Conference on Human Factors in Computing Systems - Proceedings 2019* 5;<https://dl.acm.org/doi/10.1145/3290605.3300830>.
- [180] ;.
- [181] Bird S, Dudík M, Edgar R, Horn B, Lutz R, Milan V, et al., Fairlearn: A toolkit for assessing and improving fairness in AI; 2020.
- [182] Wexler J, Pushkarna M, Bolukbasi T, Wattenberg M, Viegas F, Wilson J. The what-if tool: Interactive probing of machine learning models. *IEEE Transactions on Visualization and Computer Graphics* 2020 1;26:56–65.
- [183] Saleiro P, Kuester B, Hinkson L, London J, Stevens A, Anisfeld A, et al. Aequitas: A Bias and Fairness Audit Toolkit. *arXiv* 2018 11;.
- [184] Wisniewski J, Biecek P. fairmodels: A Flexible Tool For Bias Detection, Visualization, And Mitigation. *R Journal* 2022 4;<https://arxiv.org/abs/2104.00507v2>.
- [185] Nakao Y, Stumpf S, Ahmed S, Naseer A, Strappelli L. Toward Involving End-users in Interactive Human-in-the-loop AI Fairness. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 2022 7;12:18. <https://dl.acm.org/doi/10.1145/3514258>.
- [186] Bantilan N. Themis-ml: A Fairness-aware Machine Learning Interface for End-to-end Discrimination Discovery and Mitigation. *Journal of Technology in Human Services* 2018 10;36:15–30. <https://arxiv.org/abs/1710.06921v1>.
- [187] Purificato E, Lorenzo F, Fallucchi F, Luca EWD. The Use of Responsible Artificial Intelligence Techniques in the Context of Loan Approval Processes. *International Journal of Human-Computer Interaction* 2022;<https://www.tandfonline.com/doi/abs/10.1080/10447318.2022.2081284>.
- [188] Liu LT, Dean S, Rolf E, Simchowitz M, Hardt M. Delayed impact of fair machine learning. *IJCAI International Joint Conference on Artificial Intelligence 2019;2019-August*:6196–6200.
- [189] Puranik B, Madhoo U, Pedarsani R. A Dynamic Decision-Making Framework Promoting Long-Term Fairness. *AIES 2022 - Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society 2022* 7;p. 547–556.
- [190] D'Amour A, Srinivasan H, Atwood J, Baljekar P, Sculley D, Halpern Y. Fairness is not static: Deeper understanding of long term fairness via simulation studies. *FAT* 2020 - Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency 2020* 1;p. 525–534.
- [191] Wen M, Bastani O, Topcu U. Algorithms for Fairness in Sequential Decision Making. *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics 2021* 3;130:1144–1152. <https://proceedings.mlr.press/v130/wen21a.html>.

- [192] Saxena NA, Huang K, DeFilippis E, Radanovic G, Parkes DC, Liu Y. How do fairness definitions fare? Testing public attitudes towards three algorithmic definitions of fairness in loan allocations. *Artificial Intelligence* 2020 6;283:103238.
- [193] Gemalmaz MA, Yin M. Understanding Decision Subjects' Fairness Perceptions and Retention in Repeated Interactions with AI-Based Decision Systems. *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society* 2022; <https://doi.org/10.1145/3514094.3534201>.
- [194] Albach M, Wright JR. The Role of Accuracy in Algorithmic Process Fairness across Multiple Domains. *EC 2021 - Proceedings of the 22nd ACM Conference on Economics and Computation* 2021 7;p. 29–49.
- [195] Schoeffer J, Kuehl N, MacHowski Y. "There Is Not Enough Information": On the Effects of Explanations on Perceptions of Informational Fairness and Trustworthiness in Automated Decision-Making. *ACM International Conference Proceeding Series* 2022 6;p. 1616–1628. <https://dl.acm.org/doi/10.1145/3531146.3533218>.
- [196] Mehrabi N, Morstatter F, Saxena N, Lerman K, Galstyan A. A survey on bias and fairness in machine learning. *ACM Computing Surveys* 2019 8;54.
- [197] Yezer AM. A Review of Statistical Problems in the Measurement of Mortgage Market Discrimination and Credit Risk. *Research Institute for Housing America* 2010;.
- [198] Quillian L, Lee JJ, Honoré B. Racial Discrimination in the U.S. Housing and Mortgage Lending Markets: A Quantitative Review of Trends, 1976–2016. *Race and Social Problems* 2020 3;12:13–28. <https://link.springer.com/article/10.1007/s12552-019-09276-x>.
- [199] Yinger J. Acts of discrimination: Evidence from the 1989 housing discrimination study. *Journal of Housing Economics* 1991 12;1:318–346.
- [200] Lessmann S, Baesens B, Seow HV, Thomas LC. Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research* 2015;247:124–136.
- [201] Leo M, Sharma S, Maddulety K. Machine learning in banking risk management: A literature review. *Risks* 2019;7:29.
- [202] Mhlana D, Boubaker S. Financial Inclusion in Emerging Economies: The Application of Machine Learning and Artificial Intelligence in Credit Risk Assessment. *International Journal of Financial Studies* 2021, Vol 9, Page 39 2021 7;9:39. <https://www.mdpi.com/2227-7072/9/3/39/html> <https://www.mdpi.com/2227-7072/9/3/39>.
- [203] Khandani AE, Kim AJ, Lo AW. Consumer credit-risk models via machine-learning algorithms. *Journal of Banking Finance* 2010 11;34:2767–2787.
- [204] Heng YS, Subramanian P. A Systematic Review of Machine Learning and Explainable Artificial Intelligence (XAI) in Credit Risk Modelling. *Lecture Notes in Networks and Systems* 2023;559 LNNS:596–614. https://link.springer.com/chapter/10.1007/978-3-031-18461-1_39.
- [205] Bückner M, Szepannek G, Gosiewska A, Biecek P. Transparency, auditability, and explainability of machine learning models in credit scoring. *Journal of Operational Research Society* 2021;73:70–90. <https://www.tandfonline.com/doi/abs/10.1080/01605682.2021.1922098>.
- [206] Bueff AC, Cytryński M, Calabrese R, Jones M, Roberts J, Moore J, et al. Machine learning interpretability for a stress scenario generation in credit scoring based on counterfactuals. *Expert Systems with Applications* 2022 9;202:117271.
- [207] Crenshaw K. Demarginalizing the intersection of race and sex: A black feminist critique of antidiscrimination doctrine, feminist theory and antiracist politics. *University of Chicago Legal Forum* 1989;1989.
- [208] Creager E, Madras D, Pitassi T, Zemel R. Causal Modeling for Fairness in Dynamical Systems. *Proceedings of the 37th International Conference on Machine Learning* 2020;p. 2185.

- [209] Wachter S, Mittelstadt B, Russell C. Why fairness cannot be automated: Bridging the gap between EU non-discrimination law and AI. *Computer Law Security Review* 2021 7;41:105567.
- [210] Makhoul K, Zhioua S, Palamidessi C. Machine learning fairness notions: Bridging the gap with real-world applications. *Information Processing and Management* 2020 6;58. <http://arxiv.org/abs/2006.16745><http://dx.doi.org/10.1016/j.ipm.2021.102642>.